

INFOSYS SPRINGBOARD

VIRTUAL INTERNSHIP 7.0

Internship Completion Report

Team Details : Name - Team B

Batch Number : 01

Start Date : 29 June 2026

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Internship Duration : 8 Weeks

1. Project Title

“ Cyber Threat Analytics Platform with Incident Visualization System Group-1”

2. Project Objective

The main objectives of this project are:

- To collect and analyze cybersecurity incident data.
- To monitor security events efficiently.
- To identify suspicious activities and emerging cyber threats.
- To analyze attack trends across different countries and industries.
- To classify incidents based on attack type and severity.
- To evaluate financial losses caused by cyber attacks.
- To develop interactive dashboards for security monitoring.

3. Project Description In Detail :

3.1 Overview

The Cyber Threat Analytics Platform with Incident Visualization System is a Business Intelligence project developed using Microsoft Power BI to analyze and visualize global cybersecurity incidents. As cyber threats continue to increase across various

Countries and industries, organizations require efficient analytical tools to monitor attack patterns, assess risks, and support strategic decision-making. This project addresses that need by transforming raw cybersecurity data into meaningful and interactive dashboards.

The project uses a publicly available cybersecurity dataset collected from Kaggle, containing approximately 3,000 records covering incidents from 2015 to 2024. The dataset includes information such as Country, Year, Attack Type, Target Industry, Attack Source, Financial Loss, Affected Users, Security Vulnerability Type, Defense Mechanism Used, Resolution Time, and Risk Level.

The dataset was imported into Power BI, where Power Query Editor was used to perform data cleaning and transformation. After preprocessing, DAX measures and KPIs were created to analyze important cybersecurity metrics. Finally, five interactive dashboards were developed to present insights related to cyber attack trends, financial impact, industry risks, geographical distribution, vulnerabilities, and defense mechanisms. The project demonstrates how Business Intelligence tools can help organizations monitor cybersecurity threats and make informed decisions through data-driven insights.

3.2 Project Approach

The project was carried out using a structured eight-step approach to ensure systematic execution, accurate analysis, and effective dashboard development.

Step 1: Problem Understanding

The project began by understanding the problem statement and defining the project objectives. The goal was to analyze global cybersecurity incidents and develop interactive dashboards that provide insights into attack patterns, financial losses, affected users, industry risks, and defense mechanisms. The project scope and expected outcomes were finalized before starting the implementation.

Step 2: Data Collection

A cybersecurity dataset was collected from Kaggle. The dataset contains approximately 3,000 records covering cybersecurity incidents from 2015 to 2024. It includes attributes such as Country, Year, Attack Type, Target Industry, Attack Source, Financial Loss, Affected Users, Security Vulnerability Type, Defense Mechanism Used, Resolution Time, and Risk Level.

Step 3: Data Understanding

The collected dataset was analyzed to understand its structure, identify important attributes, and check for missing values, duplicate records, incorrect data types, and inconsistent formatting. This helped in determining the preprocessing techniques required before analysis.

Step 4: Data Connection

The dataset was imported into Microsoft Power BI Desktop using the **Get Data** feature. After establishing the connection, the data was loaded into Power Query Editor for transformation and preparation. This ensured that the dataset was ready for further processing and dashboard development.

Step 5: Data Preprocessing

Power Query Editor was used to clean and transform the dataset. Duplicate records were removed, missing values were handled, data types were corrected, unnecessary spaces were trimmed, and text values were standardized. These preprocessing steps improved the quality and consistency of the dataset, making it suitable for analysis.

Step 6: Data Modelling

After preprocessing, DAX measures and KPIs were created to calculate important metrics such as Total Financial Loss, Total Attacked Users, Average Resolution Time, Total Incidents, and High-Risk Incidents. Proper data modelling ensured accurate calculations and enabled interactive filtering across all dashboard pages.

Step 7: Data Visualization

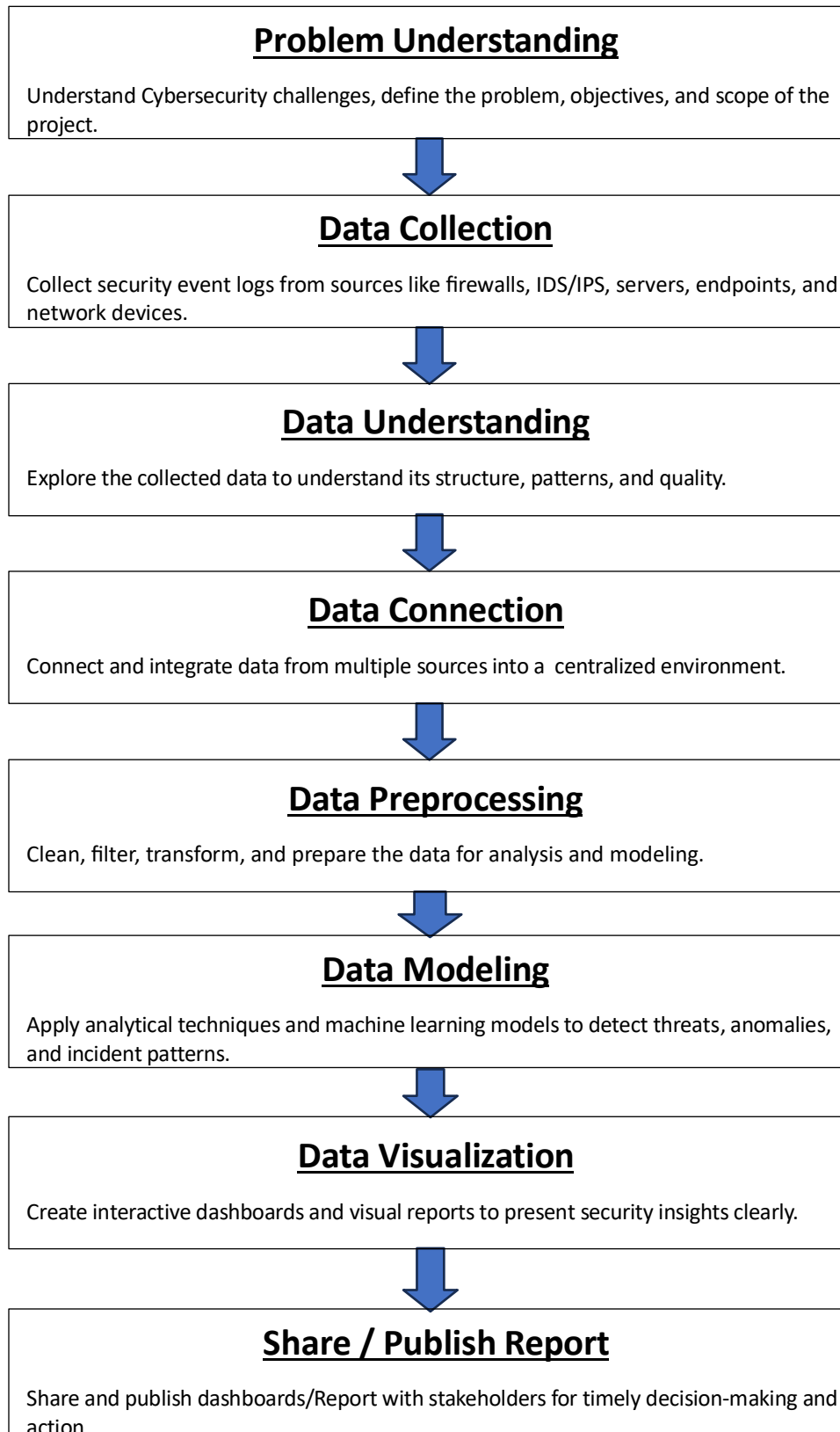
Five interactive Power BI dashboards were developed using KPI cards, bar charts, donut charts, tree maps, maps, tables, and slicers. These dashboards provided insights into cybersecurity trends, attack analysis, industry risk, country-wise analysis, and

vulnerability and defense mechanisms. Interactive filters allowed users to explore the data dynamically.

Step 8: Share / Publish Report

After completing the dashboards, the project was reviewed and validated to ensure accuracy. The final Power BI report was prepared along with the internship documentation, including dashboard screenshots, insights, and conclusions. The completed project was then submitted as part of the Infosys Springboard Virtual Internship.

Workflow Diagram



3.3 Technology Used

Technology	Purpose
Microsoft Power BI Desktop	Creating interactive dashboards and reports
Power Query Editor	Data cleaning and transformation
DAX (Data Analysis Expressions)	Creating calculated measures and KPIs
CSV Dataset	Source of cybersecurity incident data
Kaggle	Public dataset provider
Microsoft Excel	Dataset verification and review
Bing Maps	Country-wise geographical visualization
GitHub	Used for version control, source code management, and maintaining the project repository
GitLab	Used for repository management, version control, and collaboration during project development
Microsoft Teams	Used for team communication and meetings

3.4 Insights from the Dashboard

The developed dashboards provide valuable insights into global cybersecurity incidents. The **Executive Dashboard** presents an overall summary of financial losses, affected users, average resolution time, and country-wise attack distribution. The **Cyber Attack & Threat Analysis Dashboard** highlights attack trends, attack sources, attack types, and yearly incident patterns. The **Industry Risk Profile Dashboard** compares industries based on

financial losses, affected users, attack frequency, and incident resolution time. The **Country Deep Dive Dashboard** visualizes geographical attack distribution and country-wise financial losses using maps and ranking charts. The **Vulnerability & Defense Insights Dashboard** analyzes different vulnerability types, defense mechanisms, financial losses caused by vulnerabilities, and the effectiveness of cybersecurity defenses. The **Severity & Financial Impact Analysis Dashboard** focuses on high-risk incidents across industries and countries, average loss per attack source, high-risk loss ratios, and the relationship between financial loss and resolution time. The **Pattern Radar Dashboard** identifies important cybersecurity patterns through yearly loss trends, country-wise loss trends, risk-level distribution, and cumulative high-risk incidents over time. Together, these seven dashboards provide a comprehensive view of global cybersecurity threats, helping identify high-risk areas, understand attack behavior and patterns, evaluate financial impact, and support effective strategic decision-making.

3.5 Real-World Impact for Media and Public Communication

The project demonstrates the practical application of Business Intelligence in cybersecurity monitoring and public communication. Government agencies, private organizations,

cybersecurity analysts, and media professionals can use these dashboards to monitor cyber attack trends, identify vulnerable industries, and communicate cybersecurity risks effectively. The visual representation of complex data simplifies information for both technical and non-technical audiences, improving awareness and understanding of emerging cyber threats. Organizations can use these insights to strengthen security strategies, optimize resource allocation, improve incident response planning, and support data-driven decision-making. Overall, the project highlights how interactive data visualization contributes to better cybersecurity awareness and more effective communication in today's digital environment.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Problem understanding and project planning	Understood the project problem statement, finalized the project topic, and prepared the Project Proposal (PPT).
Week 2	Data collection, report preparation, and data preprocessing	Collected the cybersecurity dataset from Kaggle, cleaned and transformed the dataset using Power Query, prepared a high-quality dataset, and completed the PPT and report for Milestone 1.
Week 3	Dashboard development (Phase 1)	Developed the first dashboard pages in Power BI with KPI cards,

		charts, slicers, DAX measures, and interactive visualizations.
Week 4	Dashboard development (Phase 1) and documentation	Completed three interactive dashboards, tested the visualizations and DAX calculations, improved formatting, updated the PPT and report, and completed Milestone 2.
Week 5	Milestone 2 – Dashboard Development	Developed the 4th and 5th dashboard pages, focusing on industry-wise risks, country-wise attack distribution, financial losses, vulnerability types, and defense mechanisms.
Week 6	Milestone 3 – Dashboard Validation and Documentation	Completed and validated the five dashboards, tested slicers and chart interactions, and updated the PPT and internship report with dashboard explanations, insights, conclusions, and learnings.
Week 7	Testing, Validation and Refinement	Tested all seven dashboards, verified calculations and visual interactions, identified errors, and refined the dashboards for better accuracy and usability.
Week 8	Documentation and Final Presentation	Finalized the PPT and project report, documented the dashboard insights, conclusions, and learnings, and prepared the final project presentation.

5a. Key Milestones

Milestone	Description	Date Achieved
<i>Project Kickoff</i>	Successfully initiated the project by understanding the problem statement, defining the project objectives, selecting the project topic, and preparing the initial project proposal.	02/07/26
<i>Dataset Preparation</i>	Successfully collected the cybersecurity dataset, completed data cleaning and preprocessing using Power Query, transformed the raw data into a high-quality dataset, and completed Milestone 1.	16/07/26
<i>Dashboard Development</i>	Successfully developed the initial five interactive Power BI dashboards with DAX measures, KPI cards, charts, slicers, and visualizations, and completed Milestone 2 and Milestone 3 activities.	29/07/26
<i>Project Validation & Final Submission</i>	Successfully developed the 6th Dashboard – Severity & Financial Impact Analysis and 7th Dashboard – Pattern Radar, validated all seven dashboards, updated the PPT and final project report, and completed the final documentation and submission for Milestone 4.	19/08/26

Milestone 1 – Project Planning, Data Collection, and Data Preparation

The first milestone focused on understanding the project requirements and preparing the dataset for analysis. Initially, the problem statement was studied to identify the project objectives and expected outcomes. The project title, objectives, and implementation approach were finalized. A cybersecurity dataset containing approximately 3,000 records was collected from Kaggle. The dataset included information such as Country, Year, Attack Type, Target Industry, Attack Source, Financial Loss, Affected Users, Security Vulnerability Type, Defense Mechanism Used, Resolution Time, and Risk Level.

The collected dataset was imported into Microsoft Power BI Desktop. Using Power Query Editor, data preprocessing activities such as removing duplicate records, handling missing values, correcting data types, trimming extra spaces, and standardizing text values were performed. These steps improved the overall quality of the dataset and prepared it for dashboard development. Along with the dataset preparation, the initial PowerPoint presentation and project report were also prepared and submitted as Milestone 1.

Milestone 2 – Dashboard Development (Phase 1)

The second milestone focused on developing the initial Power BI dashboards. After preprocessing the data, DAX measures and Key Performance Indicators (KPIs) were created to calculate important business metrics such as Total Financial Loss, Total Affected Users, Average Resolution Time, Total Cyber Attacks, and High-Risk Incidents.

Using these measures, the first two dashboard pages were designed. The Executive Summary Dashboard was developed to provide an overview of cybersecurity incidents using KPI cards, bar charts, maps, and slicers. The Attack Type Analysis Dashboard was created to analyze attack trends, attack sources, attack frequency, and yearly attack distribution using interactive visualizations.

Dashboard formatting, color themes, navigation, and interactive slicers were implemented to improve usability. The PowerPoint presentation and project report were updated with dashboard screenshots, explanations, and analytical observations before submitting Milestone 2.

Milestone 3 – Dashboard Completion(Phase 2) & Documentation

The milestone focused on completing the remaining dashboards and preparing project deliverables. Three additional dashboard

pages were developed, namely the Industry Risk Profile Dashboard, Country Deep Dive Dashboard, and Vulnerability & Defense Insights Dashboard. These dashboards provided detailed analysis of industry-wise risks, country-wise attack distribution, financial losses, vulnerability types, and defense mechanisms.

After completing five dashboards, every visualization was tested to ensure accurate calculations and proper interaction between slicers and charts. Necessary formatting improvements and validations were performed to enhance the overall dashboard quality.

Finally, the complete PowerPoint presentation and internship report were updated with detailed explanations, dashboard screenshots, insights, conclusions, and learnings. All project documents and Power BI dashboards were reviewed thoroughly and submitted successfully for Milestone 3.

Milestone 4 – Advanced Dashboard Development, Validation & Final Submission

The milestone focused on developing the final two advanced dashboards and completing the overall project. Two additional dashboard pages were developed, namely the Severity & Financial Impact Analysis Dashboard and Pattern Radar

Dashboard. These dashboards provided deeper analysis of incident severity, financial impact, attack patterns, recurring trends, and relationships between different cybersecurity factors.

After completing all **seven dashboards**, every visualization was thoroughly tested to ensure accurate calculations, proper DAX results, and smooth interaction between slicers, filters, and charts. Necessary formatting improvements, data validations, and consistency checks were performed to enhance the overall quality and usability of the dashboards.

Finally, the complete **PowerPoint presentation and internship report** were updated with detailed explanations of all seven dashboards, screenshots, key insights, conclusions, and learnings. The complete project, including all Power BI dashboards and supporting documentation, was thoroughly reviewed, validated, and successfully submitted for Milestone 4.

5b. Project Execution Details

The project was executed through a systematic eight-step methodology to ensure accurate data analysis and effective dashboard development.

Problem Understanding / Problem Statement

Cyber attacks are becoming more frequent and sophisticated, making cybersecurity one of the biggest challenges faced by organizations today. Businesses generate enormous volumes of

security logs every day, making it extremely difficult to identify genuine threats through manual analysis.

Traditional monitoring methods consume significant time and resources, while hidden threats often remain unnoticed among normal security events. Delayed detection can lead to severe financial losses, data theft, and operational failures.

Organizations require a reliable analytics solution that can process large volumes of cybersecurity data efficiently, identify suspicious activities quickly, and present meaningful insights through interactive visualizations. This project addresses these challenges by leveraging Power BI to analyze cybersecurity incidents and improve security monitoring and decision-making.

Data Collection & Understanding

The dataset used for this project was collected from **Kaggle** ([Global-cybersecurity-threats-2015-2024](#)). The dataset is titled "**Global Cybersecurity Threats (2015–2024)**" and contains cybersecurity incident records collected from different countries over a period of ten years.

The dataset consists of **3,000 rows** and **10 columns**, providing comprehensive information about various cyber attacks, affected industries, security vulnerabilities, financial losses, and defense mechanisms.

Before starting the analysis, the dataset was carefully examined to understand its structure, identify important attributes, verify data types, and detect missing or duplicate values.

Dataset Overview

- Dataset Source: Kaggle
- Dataset Name: Global Cybersecurity Threats (2015–2024)
- File Format: CSV
- Total Rows: 3,000
- Total Columns: 10



Columns Description

Column Name	Column Description
Country	This column represents the country where the cyber attack occurred. It helps identify geographical trends and compare cybersecurity incidents across different regions.
Year	This column indicates the year in which the cyber attack took place. It is useful for analyzing yearly attack trends and identifying changes in cyber threat patterns over time.
Attack Type	This column specifies the type of cyber attack, such as Malware, Phishing, DDoS, SQL Injection, or Ransomware. It helps understand the most common attack techniques used by cybercriminals.
Target Industry	This column identifies the industry that was targeted during the cyber attack, such as Healthcare, Finance, Government, Retail, Technology, or Education. It helps determine which sectors experience the highest number of cyber incidents.
Attack Source	This column indicates the origin or source of the cyber attack. It provides insights into where attacks are initiated and helps identify potential threat sources.
Financial Loss (\$M)	This column represents the estimated financial loss caused by each cyber attack in millions of dollars. It enables organizations to assess the economic impact of cybersecurity incidents.
Attacked Users	This column shows the number of users affected by each cyber attack. It helps measure the overall impact of incidents on individuals and organizations.

Security Vulnerability Type	This column describes the security weakness that allowed the attack to occur, such as weak passwords, software vulnerabilities, or system misconfigurations. It helps identify common security gaps.
Defense Mechanisms Used	This column records the security measures implemented to defend against cyber attacks, such as Firewalls, Multi-Factor Authentication, Antivirus Software, Encryption, or Security Awareness Training
Resolution Time (Hours)	This column indicates the number of hours required to resolve each cybersecurity incident. It helps evaluate incident response efficiency and organizational preparedness.

IMAGE OF DATASET

Country	Year	Attack Type	Target Industry	Financial Loss	Number of Affected	Attack Source	Security Vulnerability Type	Defense Mechanism	Incident Resolution Time (in Hours)
China	2019	Phishing	Education	80.53	773169	Hacker Group	Unpatched Software	VPN	63
China	2019	Ransomware	Retail	62.19	295961	Hacker Group	Unpatched Software	Firewall	71
India	2017	Man-in-the-IT		38.65	605895	Hacker Group	Weak Passwords	VPN	20
UK	2024	Ransomware	Telecommunica	41.44	659320	Nation-state	Social Engineering	AI-based Detection	7
Germany	2018	Man-in-the-IT		74.41	810682	Insider	Social Engineering	VPN	68
Germany	2017	Man-in-the-f Retail		98.24	285201	Unknown	Social Engineering	Antivirus	25
Germany	2016	DDoS	Telecommunica	33.26	431262	Insider	Unpatched Software	VPN	34
France	2018	SQL Injector	Government	59.23	909991	Unknown	Social Engineering	Antivirus	66
India	2016	Man-in-the-f Banking		16.88	698249	Unknown	Social Engineering	VPN	47
UK	2023	DDoS	Healthcare	69.14	685927	Hacker Group	Unpatched Software	Firewall	58
China	2019	Phishing	Telecommunica	88.67	493675	Unknown	Zero-day	VPN	29
China	2016	SQL Injector	Healthcare	38.81	920768	Hacker Group	Unpatched Software	AI-based Detection	27
India	2019	Ransomware	Education	30.56	583204	Insider	Zero-day	Firewall	37
France	2023	DDoS	Healthcare	58.37	599797	Nation-state	Unpatched Software	AI-based Detection	35
France	2024	DDoS	IT	48.01	922258	Unknown	Social Engineering	Encryption	64
Australia	2022	Phishing	Banking	64.31	120789	Nation-state	Zero-day	Encryption	36
Russia	2017	Man-in-the-f Healthcare		13.04	850158	Hacker Group	Unpatched Software	AI-based Detection	67
India	2015	DDoS	IT	93.14	805278	Insider	Social Engineering	Encryption	56
UK	2019	Malware	Telecommunica	14.01	578443	Insider	Social Engineering	Firewall	28
India	2016	DDoS	IT	36.45	261808	Nation-state	Social Engineering	AI-based Detection	6
Brazil	2015	Ransomware	Retail	49.55	920172	Hacker Group	Weak Passwords	Antivirus	43
France	2023	Ransomware	Education	17.72	261661	Insider	Social Engineering	VPN	11
India	2017	DDoS	Government	76.71	246205	Nation-state	Unpatched Software	Firewall	30
Japan	2022	Phishing	Telecommunica	20.42	186148	Nation-state	Weak Passwords	Antivirus	32
Brazil	2023	DDoS	Telecommunica	51.84	699007	Hacker Group	Social Engineering	VPN	30

Data Connection

The cybersecurity dataset was downloaded from the Kaggle platform in CSV format. Microsoft Power BI Desktop was used to import and analyze the dataset.

The data connection process involved the following steps:

1. Downloaded the **Global Cybersecurity Threats (2015–2024)** dataset from Kaggle.
2. Opened Microsoft Power BI Desktop.
3. Selected **Get Data** from the Home ribbon.
4. Chose the **Text/CSV** data source.
5. Browsed and selected the downloaded CSV dataset.
6. Previewed the dataset to verify that all records and columns were correctly loaded.
7. Confirmed the delimiter and file encoding settings.
8. Clicked **Transform Data** to open the Power Query Editor.
9. Verified all column names and checked their corresponding data types.
10. Confirmed that the dataset contained 3,000 records and 10 columns.
11. Prepared the dataset for cleaning and transformation.
12. Loaded the transformed dataset into Power BI for further analysis and dashboard development.

This process ensured that the data was accurately imported and ready for preprocessing and visualization.

Data Preprocessing

Data preprocessing is an essential step that improves data quality before analysis. Several transformations were applied in Power Query Editor to clean and standardize the dataset.

The screenshot shows the Power Query Editor window with a table named 'Global_Cybersecurity_Threats'. The table contains 28 rows of data. The columns are: Country, Year, Attack Type, Target Industry, Attack Source, Financial Loss (\$M), and Affected Users. The data is sorted by Year in ascending order.

Country	Year	Attack Type	Target Industry	Attack Source	Financial Loss (\$M)	Affected Users
UK	2015	Sqli Injection	Telecommunications	Hacker Group	58.72	943070
UK	2015	Ransomware	Telecommunications	Hacker Group	22.58	368441
Japan	2015	Phishing	Education	Nation-State	58.86	587851
USA	2015	Ddos	IT	Unknown	94.14	854005
Germany	2015	Malware	Healthcare	Nation-State	90.91	654633
Brazil	2015	Phishing	Retail	Hacker Group	15.93	355008
UK	2015	Ddos	Telecommunications	Nation-State	32.92	157678
France	2015	Malware	Banking	Insider	30.15	205850
Russia	2015	Ransomware	Government	Nation-State	2.73	465465
UK	2015	Phishing	Healthcare	Unknown	81.75	98634
Russia	2015	Sqli Injection	IT	Hacker Group	47.8	645774
India	2015	Ddos	Education	Unknown	21.74	513867
France	2015	Sqli Injection	Government	Insider	14.72	540718
Australia	2015	Ddos	IT	Unknown	61.01	419127
India	2015	Phishing	Healthcare	Hacker Group	28.6	690500
Brazil	2015	Ransomware	Retail	Hacker Group	49.55	920172
India	2015	Sqli Injection	Banking	Unknown	95.81	42883
China	2015	Malware	Retail	Hacker Group	73.45	701505
Russia	2015	Malware	IT	Unknown	95.33	282598
Brazil	2015	Sqli Injection	Retail	Nation-State	74.74	562552
Australia	2015	Phishing	Healthcare	Nation-State	76.4	108767
France	2015	Ransomware	IT	Unknown	74.03	597723
India	2015	Malware	Retail	Unknown	89.95	917351
USA	2015	Man-In-The-Middle	Government	Unknown	47.76	428356
India	2015	Ransomware	Education	Hacker Group	96.03	898251
China	2015	Phishing	Government	Nation-State	42.73	310574
India	2015	Sqli Injection	IT	Unknown	42.99	85255

1. Source

Connected the original CSV dataset from the local system into Power BI.

2. Promoted Headers

Converted the first row of the dataset into column headers for proper identification of each field.

3. Changed Type

Assigned appropriate data types such as Text, Whole Number, Decimal Number, and Date where required to ensure accurate calculations.

4. Trimmed Text

Removed unnecessary leading and trailing spaces from all text columns to maintain consistency.

5. Cleaned Text

Removed non-printable and unwanted special characters from text fields, improving data quality.

6. Capitalized Each Word

Standardized text formatting by converting values into proper case, improving readability and consistency.

7. Replaced Value

Replaced inconsistent or incorrect values with standardized values to ensure uniformity throughout the dataset.

8. Removed Values

Removed unnecessary or irrelevant values that were not useful for analysis.

9. Removed Duplicates

Identified and removed duplicate records to avoid inaccurate analysis and repeated information.

10. Removed Errors

Detected and removed rows containing data errors to improve the reliability of the dataset.

11. Removed Blank Rows

Deleted empty rows that contained no meaningful information, ensuring a clean dataset.

12. Renamed Columns

Renamed column headers with meaningful and user-friendly names for better understanding.

13. Reordered Columns

Rearranged the columns into a logical sequence to improve organization and simplify data analysis.

14. Sorted Rows

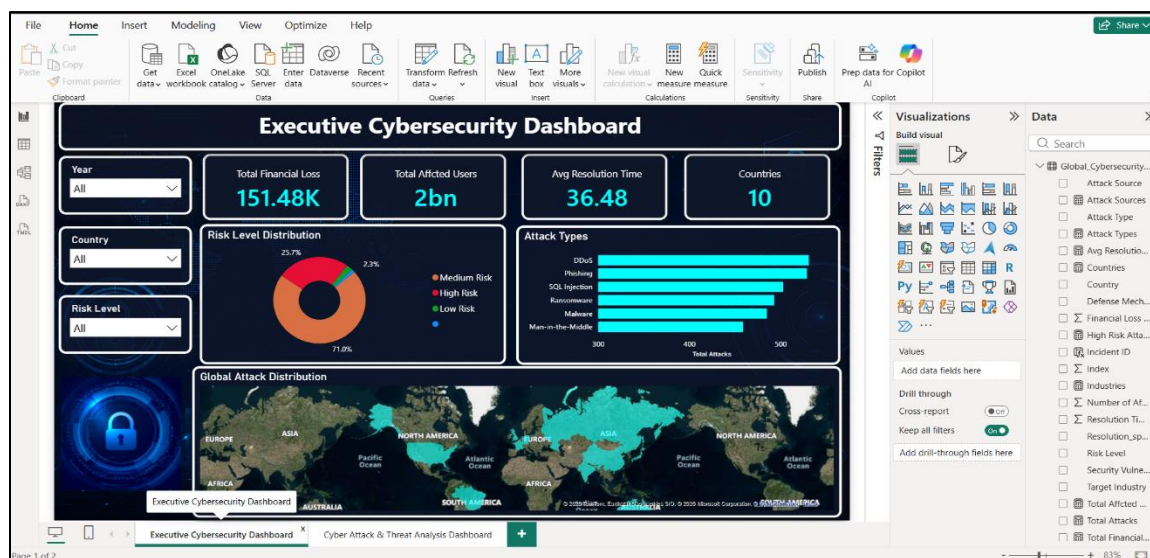
Sorted the dataset based on relevant columns to facilitate easier analysis and reporting. After completing these preprocessing steps, the dataset became clean, consistent, accurate, and ready for data modeling and visualization.

Data Visualization

Data visualization is the process of converting raw data into graphical representations such as charts, graphs, maps, and dashboards. In this project, **Microsoft Power BI** was used to create interactive dashboards that present cybersecurity incident

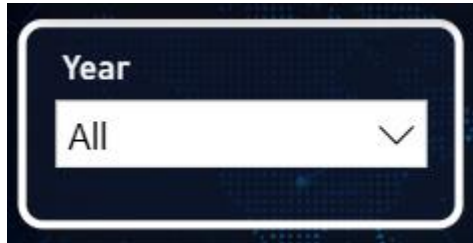
data in a clear and meaningful way. The visualizations help analyze cyber attack trends, financial losses, affected users, resolution time, and country-wise incident distribution. These dashboards enable users to identify patterns, compare different attack types, monitor cybersecurity performance, and support faster and more effective decision-making.

Executive Cybersecurity Dashboard



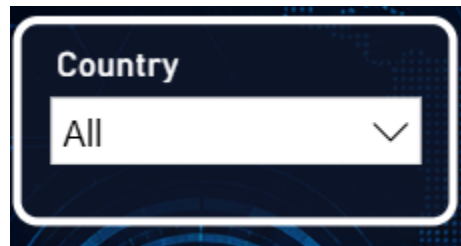
The Executive Cybersecurity Dashboard provides an overall summary of cybersecurity incidents through key performance indicators (KPIs) and interactive visualizations. It helps management quickly monitor cyber threats, financial impact, affected users, and risk distribution.

Year Slicer



The Year slicer allows users to filter the dashboard based on a selected year. When a specific year is chosen, all charts, cards, and maps update automatically to display data for that year only. This helps analyze cybersecurity incidents over different years.

Country Slicer



The Country slicer filters all dashboard visualizations according to the selected country. It enables users to compare cybersecurity incidents and financial losses across different countries.

Risk Level Slicer



The Risk Level slicer filters data based on High Risk, Medium Risk, and Low Risk incidents. This helps users focus on a particular risk category and analyze its impact separately.

Total Financial Loss (Card Visual)



This card displays the total financial loss caused by cybersecurity incidents. It provides a quick overview of the economic impact of cyber attacks and helps organizations understand the overall financial damage.

Total Affected Users (Card Visual)



This card shows the total number of users affected by cyber attacks. It indicates the scale of cybersecurity incidents and helps measure their overall impact on users.

Average Resolution Time (Card Visual)



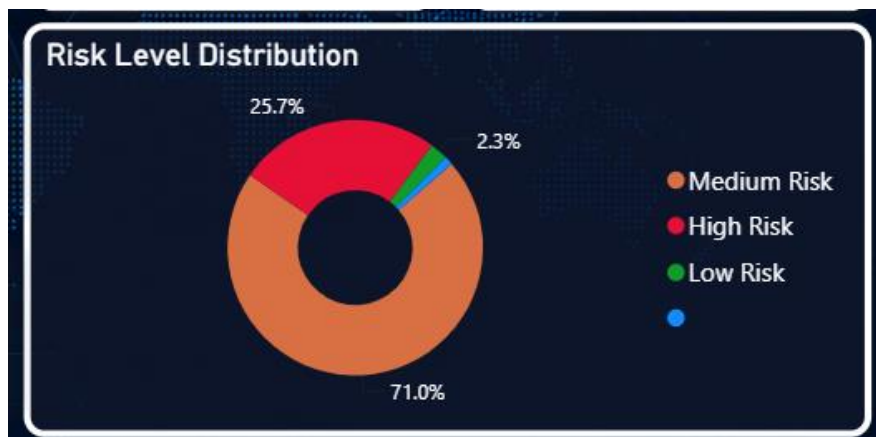
This card displays the average time required to resolve cybersecurity incidents. It helps evaluate the efficiency of incident response teams, where a lower resolution time indicates better performance.

Countries (Card Visual)



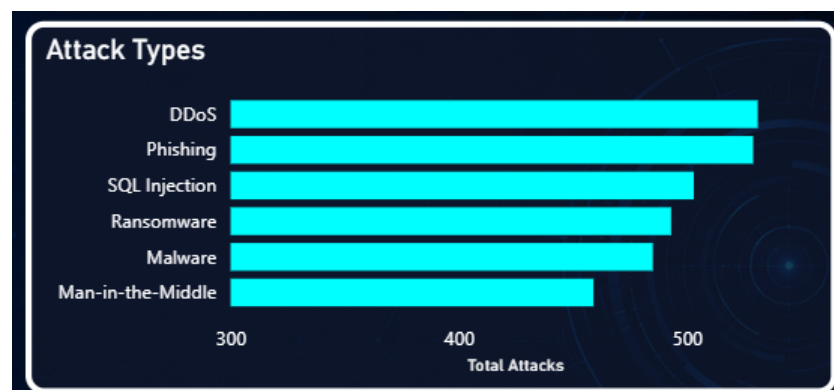
This card displays the total number of countries included in the dataset. It provides an overview of the geographical coverage of the cybersecurity analysis.

Risk Level Distribution (Donut Chart)



The donut chart represents the percentage distribution of High Risk, Medium Risk, and Low Risk incidents. It helps identify which risk level contributes the most to the total cybersecurity incidents.

Attack Type (Horizontal Bar Charts)



This chart compares the total number of incidents for each cyber attack type. It helps identify the most frequent attacks and supports prioritizing cybersecurity measures.

Global Attack Distribution (Map Visual)



The map visual displays the geographical distribution of cybersecurity incidents across different countries. It helps identify regions with higher cyber attack activity and provides a global view of cybersecurity threats.

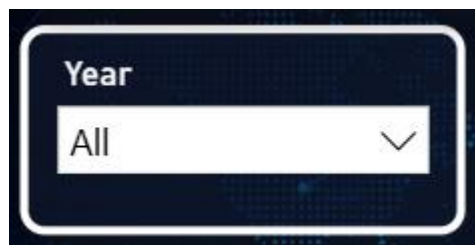
Outcome: This dashboard provides an executive-level summary of cybersecurity performance, helping organizations monitor incidents, assess risks, and make informed decisions.

Cyber Attack & Threat Analysis Dashboard



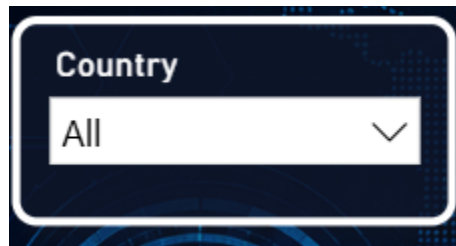
The **Cyber Attack & Threat Analysis Dashboard** provides detailed insights into cyber attack trends, attack sources, risk levels, and attack categories. It helps security analysts understand attack patterns and improve cybersecurity strategies.

Year Slicer



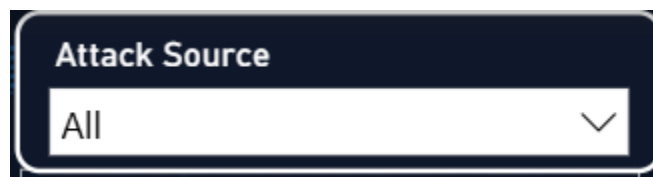
The Year slicer filters all dashboard data based on the selected year. It helps analyze how cyber attacks have changed over time.

Country Slicer



The Country slicer allows users to view attack information for a selected country. It supports country-wise comparison of cybersecurity incidents.

Attack Source Slicer



The Attack Source slicer filters incidents based on different attack sources such as Nation-State, Insider, Hacker Group, and Unknown. It helps analyze attacks according to their origin.

Risk Level Slicer



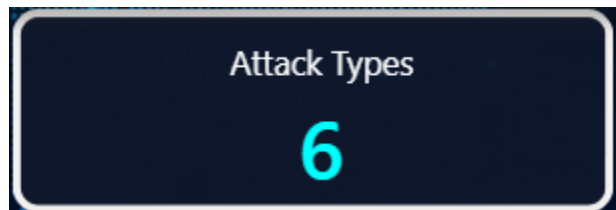
The Risk Level slicer filters the dashboard based on High Risk, Medium Risk, and Low Risk incidents, allowing users to focus on specific risk categories.

Total Attacks (Card Visual)



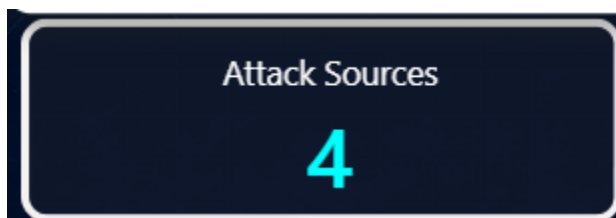
This card displays the total number of cyber attacks recorded in the dataset. It provides a quick summary of the total incidents analyzed.

Attack Types (Card Visual)



This card shows the total number of different attack categories available in the dataset. It provides an overview of the diversity of cyber attacks.

Attack Sources (Card Visual)



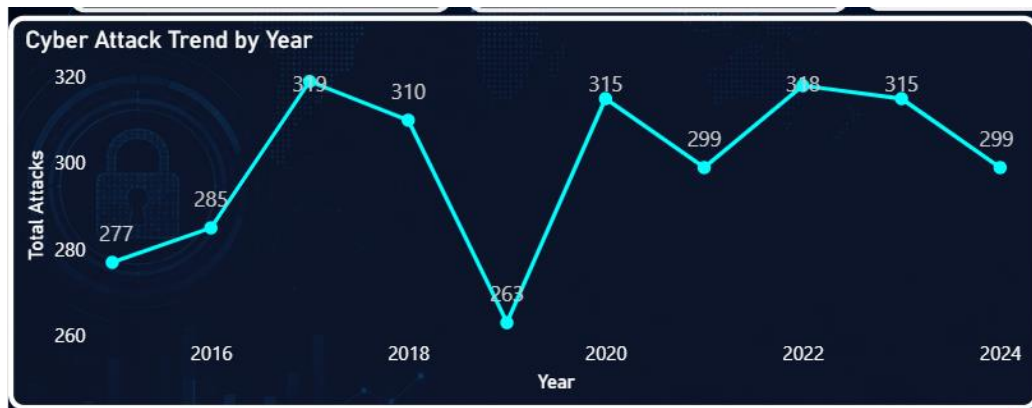
This card displays the total number of attack source categories included in the analysis. It helps users understand the different origins of cyber attacks.

High Risk Attacks (Card Visual)



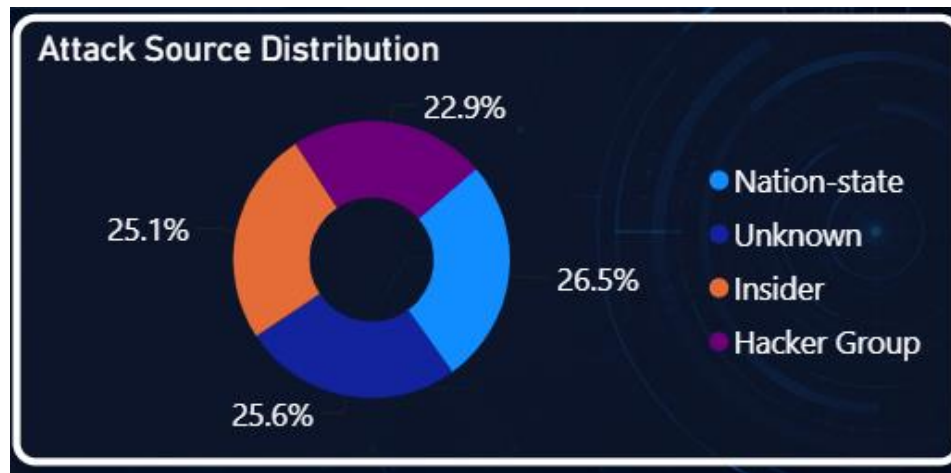
This card shows the total number of incidents classified as High Risk. It helps organizations identify critical attacks that require immediate attention.

Cyber Attack Trend by Year (Line Chart)



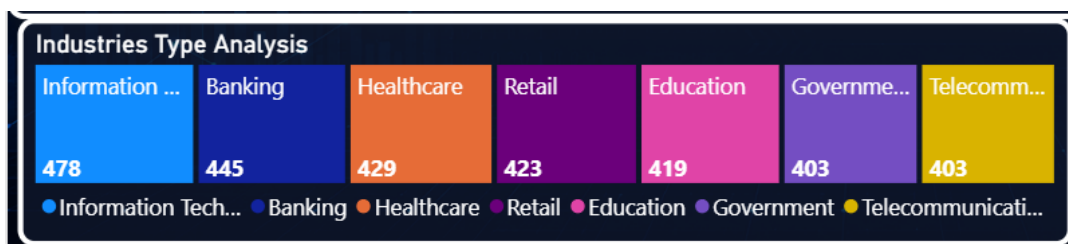
The line chart displays the yearly trend of cybersecurity incidents. It helps identify whether the number of attacks is increasing or decreasing over time and supports long-term security planning.

Attack Source Distribution (Donut Chart)



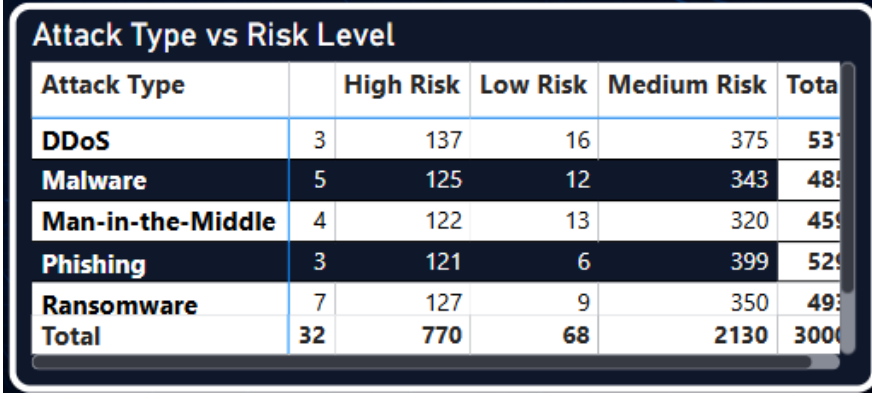
The donut chart shows the percentage distribution of different attack sources such as Nation-State, Insider, Hacker Group, and Unknown. It helps determine the most common origin of cyber attacks.

Industries Type Analysis (Treemap)



The treemap displays the distribution of cybersecurity incidents across different industries, such as Information Technology, Banking, Healthcare, Retail, Education, Government, and Telecommunications. It helps identify which industries are most frequently targeted by cyberattacks and supports better security planning and risk assessment.

Attack Type vs Risk Level (Matrix/Table Visual)



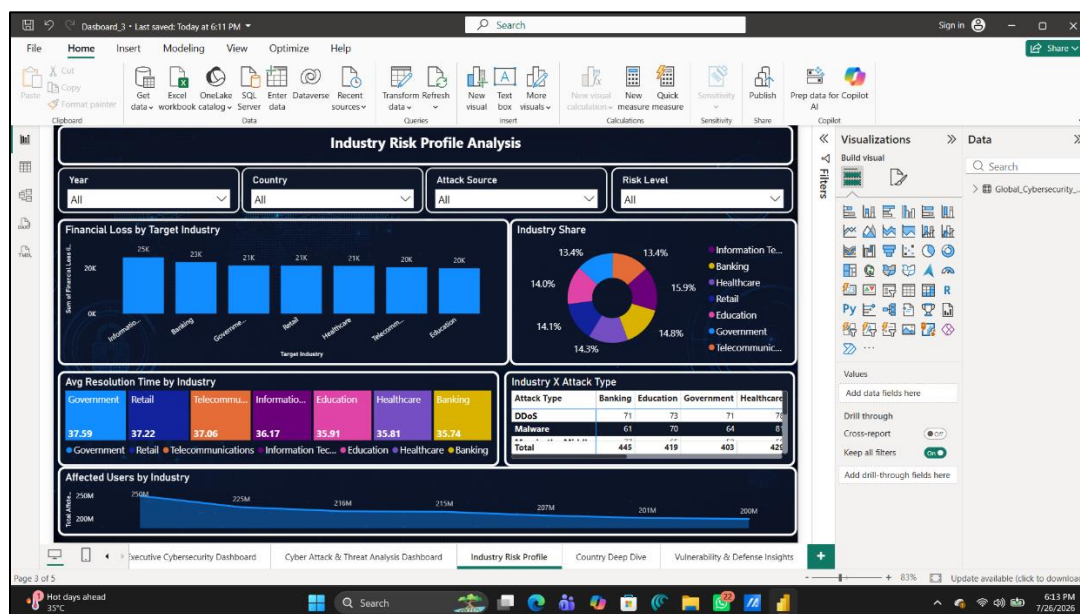
Attack Type		High Risk	Low Risk	Medium Risk	Total
DDoS	3	137	16	375	537
Malware	5	125	12	343	485
Man-in-the-Middle	4	122	13	320	459
Phishing	3	121	6	399	529
Ransomware	7	127	9	350	493
Total	32	770	68	2130	3000

The matrix visual compares each attack type with High Risk, Medium Risk, and Low Risk incidents. It provides a detailed breakdown of attack severity and helps identify which attack types pose the greatest risk to organizations.

Outcome: This dashboard provides detailed insights into cyber attack trends, attack sources, and risk levels, enabling organizations to strengthen their cybersecurity strategies and improve incident response.

Industry Risk Profile Analysis

The Industry Risk Profile Analysis dashboard provides a comprehensive analysis of how cyber attacks affect different industries. It compares industries based on financial loss, attack frequency, affected users, and incident resolution time. The dashboard helps organizations identify industries that are more vulnerable to cyber threats and supports decision-making for improving cybersecurity infrastructure. By analyzing these metrics, businesses can allocate resources efficiently, strengthen security policies, and reduce the overall impact of cyber attacks.



Year (Slider)

The Year slider allows users to filter the dashboard based on a specific year. This helps compare industry-wise cyber attack

trends over different years and identify whether the number of incidents, financial losses, or response times are increasing or

decreasing. It also enables organizations to evaluate the effectiveness of cybersecurity improvements over time.

A dark blue rectangular button with rounded corners. At the top, the word "Year" is written in white. Below it is a white dropdown menu with the word "All" in black. A small white chevron icon is on the right side of the dropdown.

Country (Slicer)

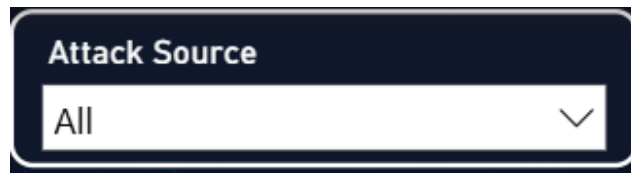
The Country slicer filters the dashboard based on a selected country. This enables users to compare cybersecurity performance across different geographical regions and understand how cyber attacks affect industries in various countries. It also helps identify countries that require stronger cybersecurity policies.

A dark blue rectangular button with rounded corners. At the top, the word "Country" is written in white. Below it is a white dropdown menu with the word "All" in black. A small white chevron icon is on the right side of the dropdown.

Attack Source (Slicer)

The Attack Source slicer categorizes incidents based on their origin, such as Insider Threats, Nation-State Attacks, Hacker Groups, or Unknown Sources. This filter allows users to analyze

which attack sources contribute most to cyber incidents within different industries.

A dark blue rectangular button with rounded corners. At the top, the text "Attack Source" is written in white. Below it is a white dropdown menu with the word "All" and a downward-pointing chevron icon on the right.

Risk Level (Slicer)

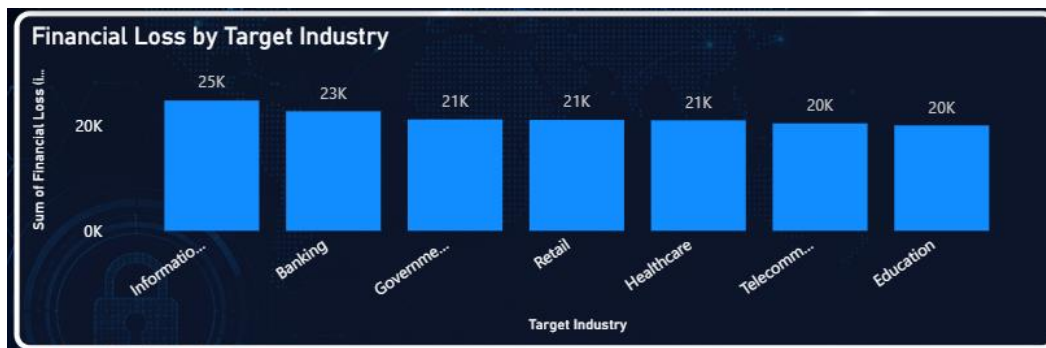
The Risk Level slicer classifies cyber incidents into Low, Medium, and High risk categories. This helps users focus on high-risk incidents and evaluate the severity of cyber attacks across industries.

A dark blue rectangular button with rounded corners. At the top, the text "Risk Level" is written in white. Below it is a white dropdown menu with the word "All" and a downward-pointing chevron icon on the right.

Financial Loss by Target Industry (Bar Chart)

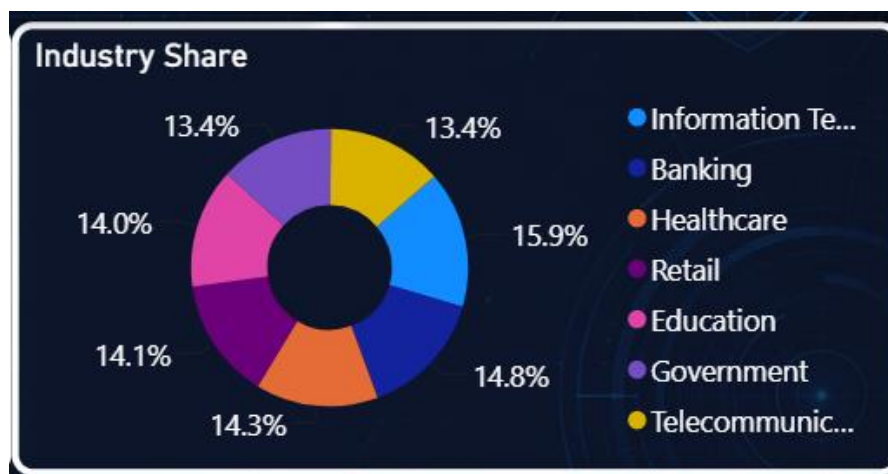
The **Bar Chart** displays the total financial loss experienced by different industries due to cyber attacks. Each bar represents an industry such as Banking, Healthcare, Government, Retail, Education, Telecommunications, and Information Technology. By comparing the height of each bar, users can easily identify which industries suffer the highest financial losses. This visualization helps organizations understand the economic impact of cyber attacks and prioritize investments in cybersecurity infrastructure. Industries with higher losses should strengthen their security measures, improve employee awareness, and

implement advanced threat detection systems to reduce future risks.



Industry Share (Donut Chart)

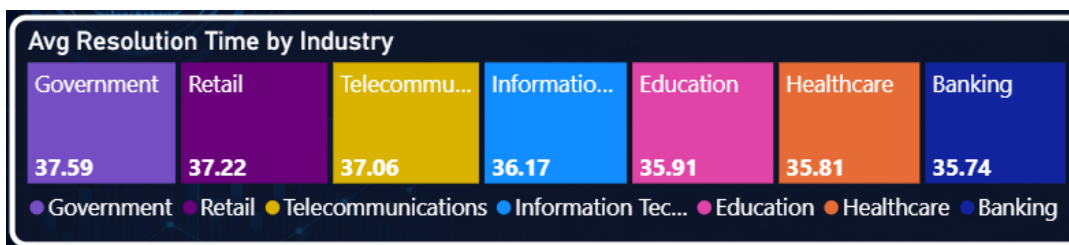
The **Donut Chart** illustrates the percentage distribution of cyber incidents across different industries. Each segment represents an industry's contribution to the total number of recorded cyber attacks. This visualization provides a quick overview of which industries are most frequently targeted by attackers. Organizations can use this information to identify sectors that require continuous monitoring and stronger cybersecurity controls. The chart also supports strategic planning by highlighting industries where security awareness programs and preventive measures should be prioritized.



Average Resolution Time by Industry (Tree map)

The **Tree map** visualizes the average time required to resolve cyber incidents across various industries. Larger blocks indicate industries with longer incident resolution times, while smaller blocks represent industries with quicker recovery. This visualization helps organizations evaluate the effectiveness of their incident response process and identify operational

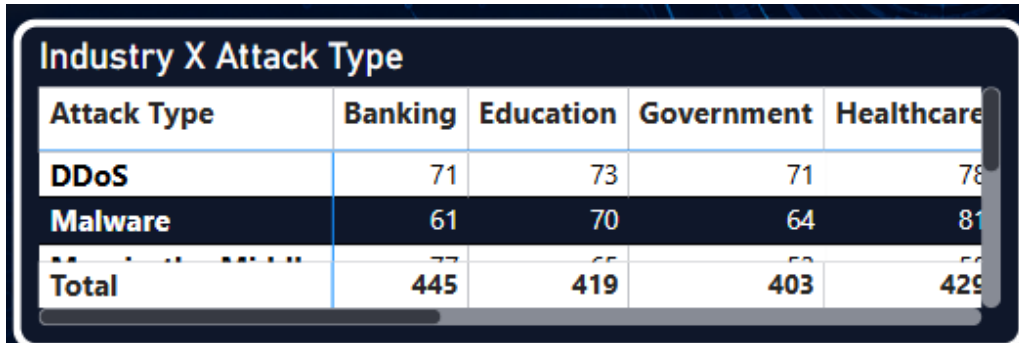
inefficiencies. Industries with higher resolution times should improve their incident response teams, automate security operations, and enhance monitoring systems to minimize business disruption.



Industry × Attack Type (Matrix Table)

The **Matrix Table** compares different cyber attack types across multiple industries. It provides detailed information about how frequently attacks such as Malware, Phishing, Ransomware, SQL Injection, and DDoS occur within each industry. This visualization helps organizations understand industry-specific attack patterns and supports the development of targeted cybersecurity strategies. It also enables security teams to allocate

resources more effectively by focusing on the most common threats affecting their industry.



Attack Type	Banking	Education	Government	Healthcare
DDoS	71	73	71	78
Malware	61	70	64	81
Total	445	419	403	429

Affected Users by Industry (Area Chart)

The **Area Chart** displays the total number of users affected by cyber attacks across different industries. It illustrates how cyber incidents impact customers, employees, and organizations over time. Industries with a larger affected user base require stronger data protection mechanisms, improved access controls, and faster incident recovery processes. This visualization helps organizations measure the real-world impact of cyber attacks and prioritize user safety and business continuity.

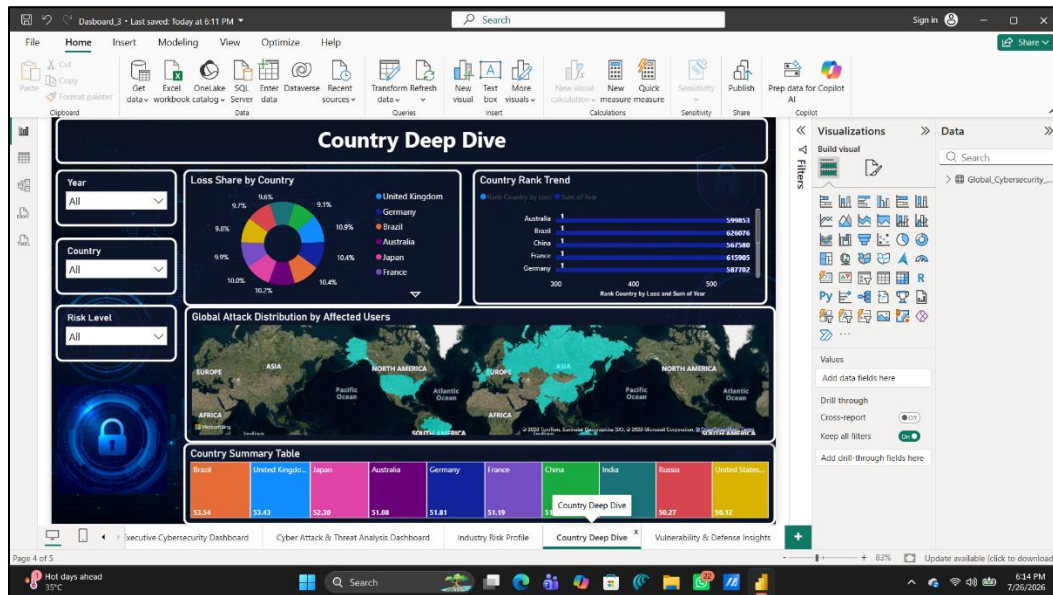


Outcome: This dashboard provides detailed country-wise insights into cyber attack distribution, financial losses, and affected users. It enables organizations and policymakers to identify high-risk regions, understand global cybersecurity trends, allocate security resources effectively, and strengthen regional cybersecurity strategies.

Country Deep Dive

The Country Deep Dive dashboard provides a comprehensive analysis of cyber attack activities across different countries. It compares countries based on financial losses, affected users, cyber attack distribution, and overall cybersecurity impact. The dashboard helps organizations and government agencies identify high-risk regions, understand country-wise cybersecurity trends, and support strategic decision-making for improving national and organizational cybersecurity. By analyzing these metrics, stakeholders can allocate security resources effectively, strengthen regional cybersecurity policies, and minimize the

impact of cyber threats across different geographical locations.



Year (Slider)

The Year slider allows users to filter the dashboard based on a specific year. This helps compare industry-wise cyber attack trends over different years and identify whether the number of

incidents, financial losses, or response times are increasing or decreasing. It also enables organizations to evaluate the effectiveness of cybersecurity improvements over time.



Country (Slicer)

The Country slicer filters the dashboard based on a selected country. This enables users to compare cybersecurity performance across different geographical regions and understand how cyber attacks affect industries in various countries. It also helps identify countries that require stronger cybersecurity policies.



Risk Level (Slicer)

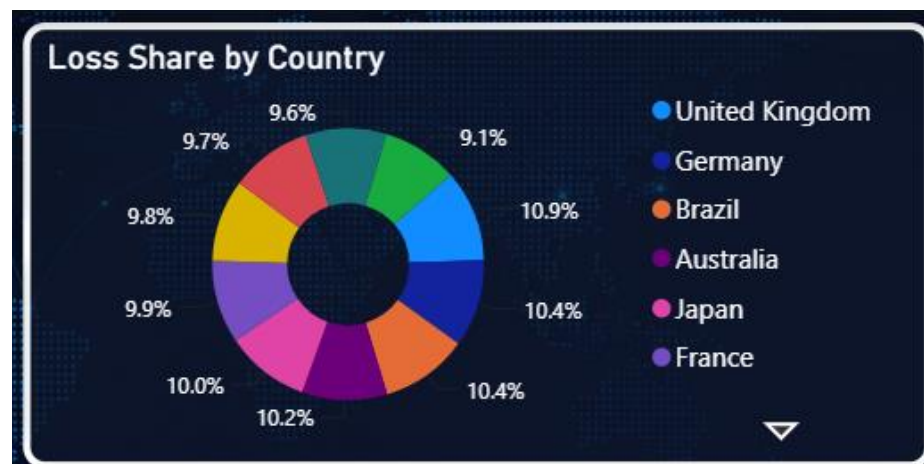
The Risk Level slicer classifies cyber incidents into Low, Medium, and High risk categories. This helps users focus on high-risk incidents and evaluate the severity of cyber attacks across industries.



Loss Share by Country (Donut Chart)

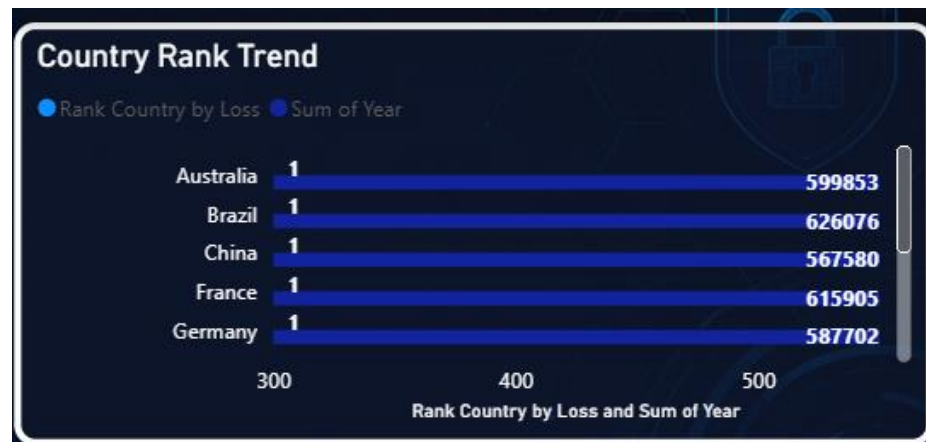
The **Donut Chart** represents the percentage contribution of each country to the total financial loss caused by cyber attacks. Each section of the chart indicates how much financial damage a country has experienced compared to others. This visualization

enables organizations and policymakers to identify countries with greater economic impact and supports better cybersecurity investment planning.



Country Rank Trend (Horizontal Bar Chart)

The **Horizontal Bar Chart** ranks countries according to the total financial losses caused by cyberattacks. Countries appearing at the top of the chart experience greater cyber risk and economic damage. This visualization simplifies country-wise comparison and helps governments and organizations prioritize cybersecurity improvements in high-risk regions.



Global Attack Distribution by Affected Users (Map)

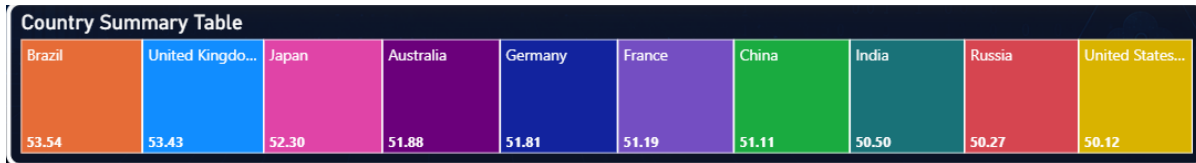
The **Map Visualization** displays the geographical distribution of cyber attacks based on the number of affected users. Countries with higher user impact are highlighted on the map, making it easy to identify global cyber attack hotspots. This visualization supports regional risk assessment and helps organizations understand where stronger cybersecurity measures are required.



Country Summary (Tree map)

The **Tree map** summarizes the overall cyber attack impact for different countries using proportional rectangles. Larger

rectangles indicate countries experiencing greater cybersecurity impact. This visualization provides an easy comparison of regional cyber risks and supports strategic cybersecurity planning.



A horizontal bar chart titled 'Country Summary Table' displaying cybersecurity impact scores for ten countries. Each country is represented by a colored rectangle with its name at the top and its score at the bottom. The scores are: Brazil (53.54), United Kingdom (53.43), Japan (52.30), Australia (51.88), Germany (51.81), France (51.19), China (51.11), India (50.50), Russia (50.27), and United States (50.12). The bars are arranged in descending order of score from left to right.

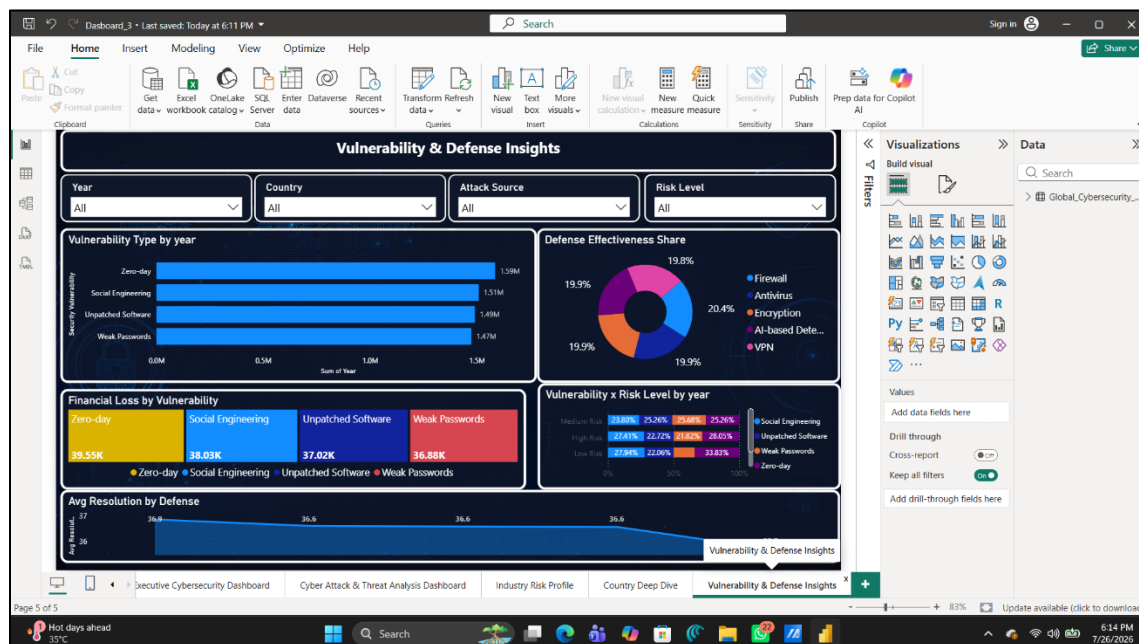
Country	Score
Brazil	53.54
United Kingdom	53.43
Japan	52.30
Australia	51.88
Germany	51.81
France	51.19
China	51.11
India	50.50
Russia	50.27
United States	50.12

Outcome: This dashboard provides detailed country-wise insights into cyber attack distribution, financial losses, and affected users. It enables organizations and policymakers to identify high-risk regions, understand global cybersecurity trends, allocate security resources effectively, and strengthen regional cybersecurity strategies.

Vulnerability & Defense Insights

The Vulnerability & Defense Insights dashboard provides a comprehensive analysis of common cybersecurity vulnerabilities and the effectiveness of various defense mechanisms. It evaluates vulnerability trends, financial losses, risk levels, and incident resolution performance to identify critical security weaknesses and assess the efficiency of existing security controls. The

dashboard helps organizations understand emerging cyber threats, improve vulnerability management, and enhance their overall cybersecurity posture. By analyzing these insights, organizations can implement stronger defense strategies, optimize incident response processes, and reduce the financial and operational impact of cyber attacks.



Year (Slicer)

The Year slicer allows users to filter the dashboard based on a specific year. This helps compare industry-wise cyber attack trends over different years and identify whether the number of incidents, financial losses, or response times are increasing or decreasing. It also enables organizations to evaluate the effectiveness of cybersecurity improvements over time.

A dark blue rectangular button with rounded corners. At the top, the word "Year" is written in white. Below it is a white dropdown menu with the word "All" and a small downward-pointing chevron icon on the right.

Country (Slicer)

The Country slicer filters the dashboard based on a selected country. This enables users to compare cybersecurity performance across different geographical regions and understand how cyber attacks affect industries in various countries. It also helps identify countries that require stronger cybersecurity policies.

A dark blue rectangular button with rounded corners. At the top, the word "Country" is written in white. Below it is a white dropdown menu with the word "All" and a small downward-pointing chevron icon on the right.

Attack Source (Slicer)

The Attack Source slicer categorizes incidents based on their origin, such as Insider Threats, Nation-State Attacks, Hacker Groups, or Unknown Sources. This filter allows users to analyze which attack sources contribute most to cyber incidents within different industries.

A dark blue rectangular button with rounded corners. At the top, the text "Attack Source" is written in white. Below it is a white dropdown menu with the word "All" and a downward-pointing chevron icon on the right.

Risk Level (Slicer)

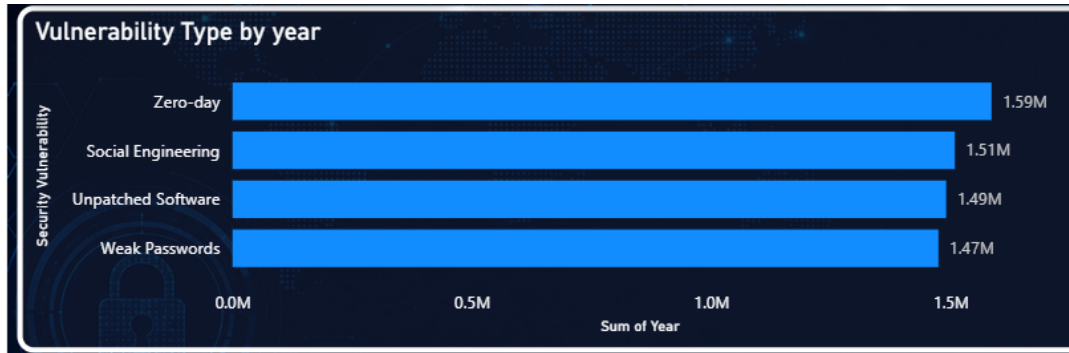
The Risk Level slicer classifies cyber incidents into Low, Medium, and High risk categories. This helps users focus on high-risk incidents and evaluate the severity of cyber attacks across industries.

A dark blue rectangular button with rounded corners. At the top, the text "Risk Level" is written in white. Below it is a white dropdown menu with the word "All" and a downward-pointing chevron icon on the right.

Vulnerability Type by Year (Horizontal Bar Chart)

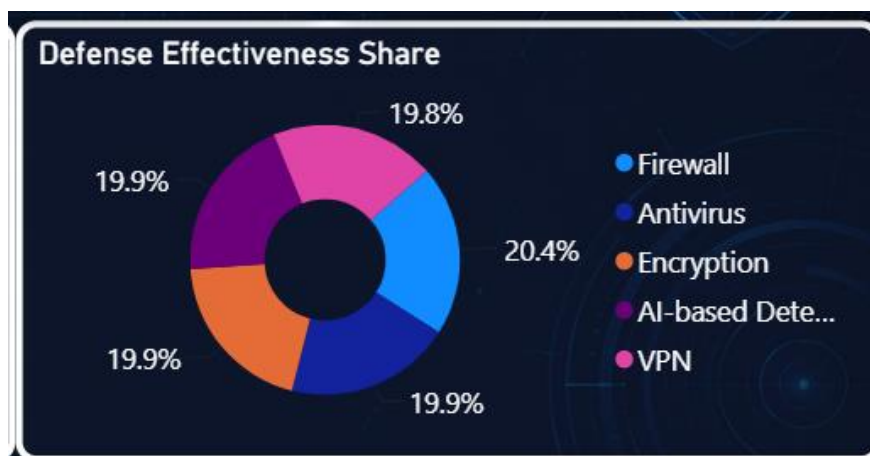
The **Horizontal Bar Chart** compares the occurrence of different vulnerability types such as Zero-day Attacks, Social Engineering, Weak Passwords, and Unpatched Software. It helps organizations identify the most common security weaknesses and prioritize vulnerability management activities. By understanding vulnerability trends, organizations can improve patch

management and strengthen security awareness programs.



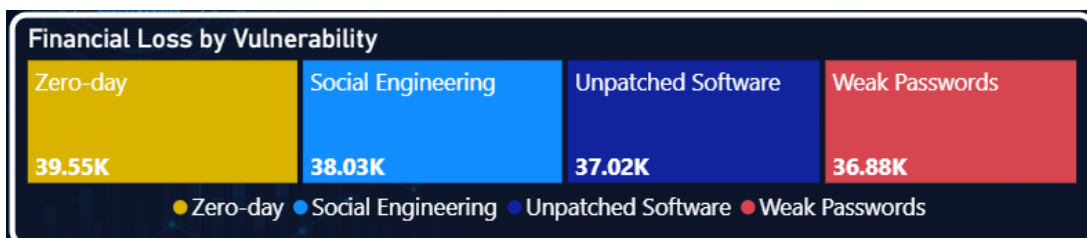
Defense Effectiveness Share (Donut Chart)

The **Donut Chart** illustrates the percentage contribution of different cybersecurity defense mechanisms including Firewalls, Antivirus Software, Encryption, AI-Based Detection Systems, and VPN technologies. It provides an overview of how different security controls contribute to organizational protection and helps evaluate the effectiveness of existing defense.



Financial Loss by Vulnerability (Tree map)

The **Tree map** compares the financial losses associated with different vulnerability types. Larger sections represent vulnerabilities responsible for greater financial damage. This visualization enables organizations to prioritize vulnerability remediation efforts and allocate cybersecurity resources more efficiently.



Vulnerability × Risk Level (100% Stacked Bar Chart)

The **100% Stacked Bar Chart** compares different vulnerability types across Low, Medium, and High risk categories. It shows the percentage distribution of vulnerabilities within each risk level, helping organizations identify which vulnerabilities contribute most significantly to severe cyber incidents. This analysis supports effective risk assessment and mitigation planning.



Average Resolution by Defense (Line Chart)

The **Line Chart** displays the average time required to resolve cyber incidents using different defense mechanisms. It compares the performance of various security technologies and helps evaluate their effectiveness in incident response. Defense mechanisms with shorter resolution times indicate better operational efficiency and stronger cybersecurity capabilities.

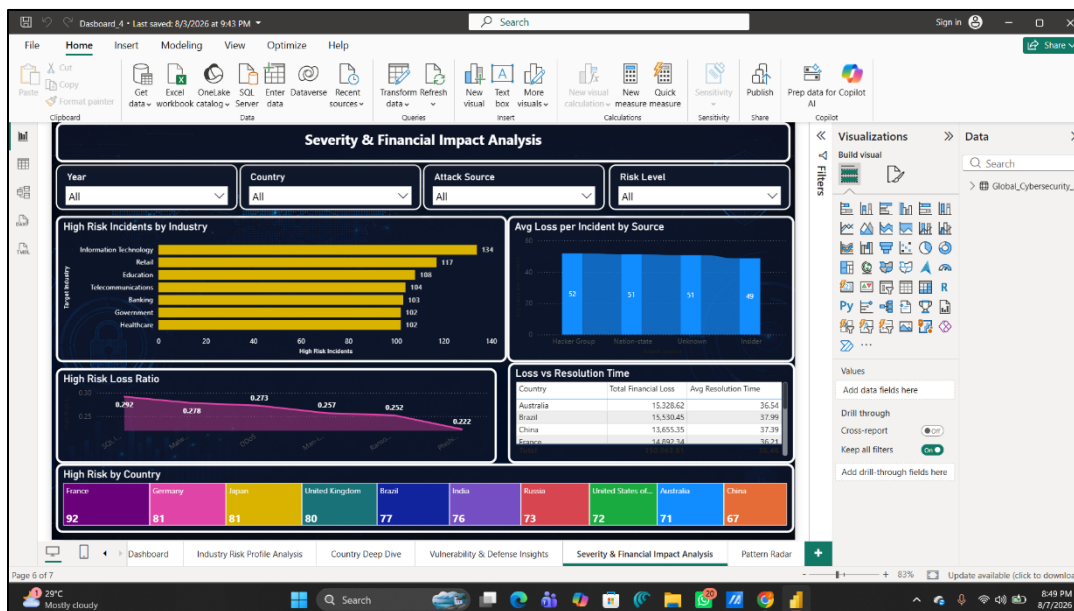


Outcome: This dashboard provides valuable insights into common cybersecurity vulnerabilities, defense mechanisms, financial losses, and incident resolution performance. It helps organizations identify critical security weaknesses, evaluate the effectiveness of security controls, prioritize vulnerability management, and enhance their overall cybersecurity resilience through informed decision-making.

Severity & Financial Impact Analysis

The **Severity & Financial Impact Analysis Dashboard** provides a comprehensive analysis of high-risk cybersecurity incidents, financial losses, attack sources, affected countries, and incident

resolution performance. It evaluates the severity of cyber incidents and their financial impact across different industries and countries. The dashboard helps organizations identify industries facing the highest number of high-risk incidents, understand which attack sources generate greater financial losses, compare high-risk loss ratios, and analyze the relationship between financial loss and resolution time. By combining severity and financial impact indicators, the dashboard supports organizations in prioritizing high-risk areas, improving incident response, and reducing the financial impact of cybersecurity threats.



Year (Slider)

Year

All

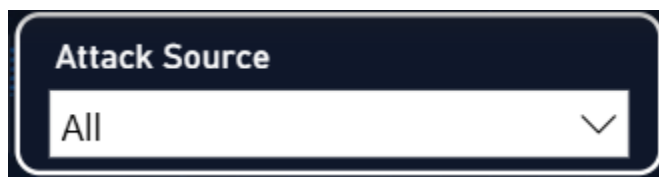
The **Year slicer** allows users to filter the dashboard based on a selected year. This helps analyze how high-risk incidents and financial losses change over time and enables users to identify years with increased cybersecurity risks.

Country (Slicer)

A dark blue rectangular button with rounded corners. At the top, the word "Country" is written in white. Below it is a white dropdown menu with the word "All" and a downward-pointing chevron icon on the right.

The **Country slicer** filters the dashboard based on a selected country. It allows users to analyze high-risk incidents and financial impact for a particular geographical region and compare cybersecurity risk across countries.

Attack Source (Slicer)

A dark blue rectangular button with rounded corners. At the top, the words "Attack Source" are written in white. Below it is a white dropdown menu with the word "All" and a downward-pointing chevron icon on the right.

The **Attack Source slicer** allows users to filter incidents based on their source, such as Hacker Groups, Nation-State Attacks, Insider Threats, or Unknown Sources. This helps identify which sources are associated with higher-risk incidents and greater financial losses.

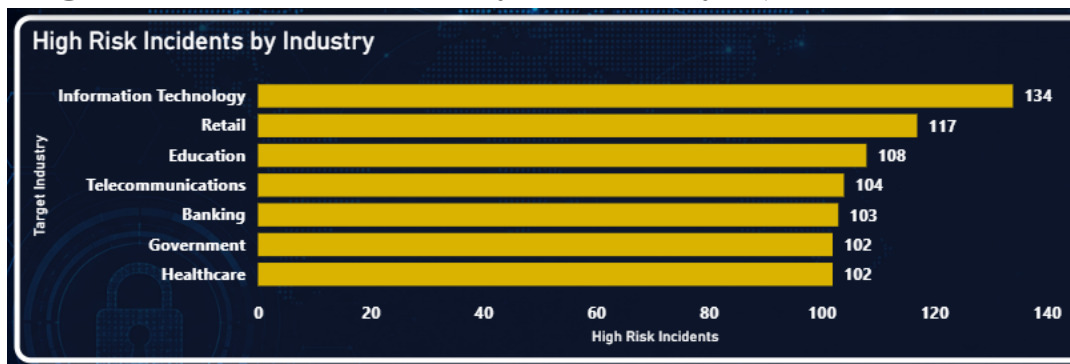
Risk Level (Slicer)

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Risk Level
All

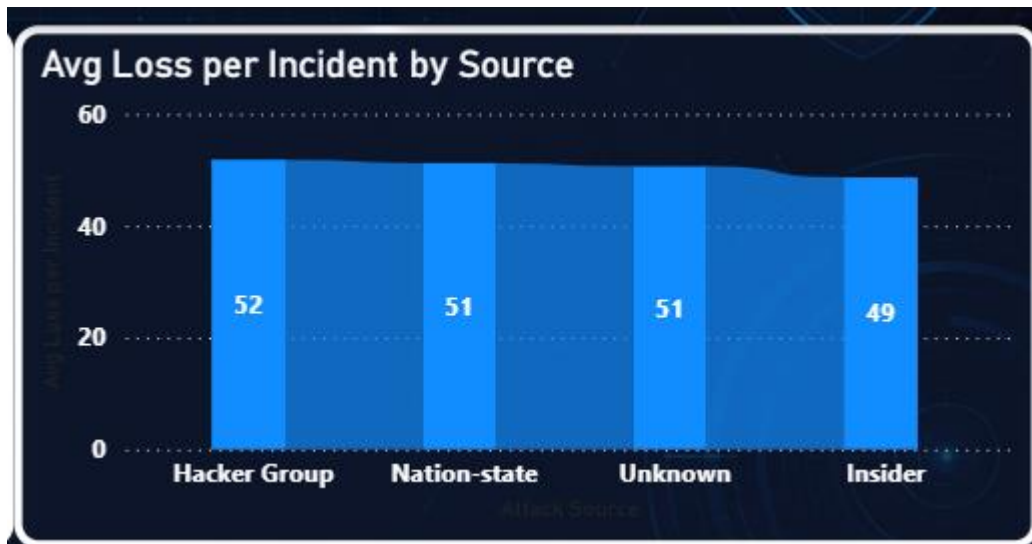
The **Risk Level slicer** filters incidents according to Low, Medium, and High risk categories. It allows users to focus specifically on high-severity incidents and evaluate their financial and operational impact.

High Risk Incidents by Industry (Horizontal Bar Chart)



The **Horizontal Bar Chart** compares the number of high-risk cybersecurity incidents across different industries such as Information Technology, Retail, Education, Telecommunications, Banking, Government, and Healthcare. It helps identify industries that experience a greater concentration of severe cyber incidents and enables organizations to prioritize security measures for high-risk sectors.

Avg Loss per Incident by Source (Column Chart)



The **Column Chart** compares the average financial loss per cybersecurity incident across different attack sources. It helps identify which sources are associated with higher financial damage. This analysis enables organizations to understand the financial severity of different attack origins and allocate resources toward stronger preventive controls.

High Risk Loss Ratio (Line Chart)



The **Line Chart** displays the high-risk loss ratio across different attack types. It shows how much financial impact is associated with high-risk incidents for each attack category. This helps

organizations identify attack types that generate comparatively higher losses and prioritize them for risk mitigation.

Loss vs Resolution Time (Table)



Country	Total Financial Loss	Avg Resolution Time
Australia	15,328.62	36.54
Brazil	15,530.45	37.99
China	13,655.35	37.39
France	14,892.34	36.21
Total	150,063.61	36.46

The **Table** compares the total financial loss and average resolution time for different countries. It helps users understand whether countries experiencing higher financial losses also require more time to resolve incidents. This supports the evaluation of incident response efficiency and helps identify regions that may require faster response mechanisms.

High Risk by Country (Treemap)



The **Treemap** visualizes the number of high-risk incidents across countries. The size of each section represents the number of high-risk incidents, allowing users to quickly identify countries with

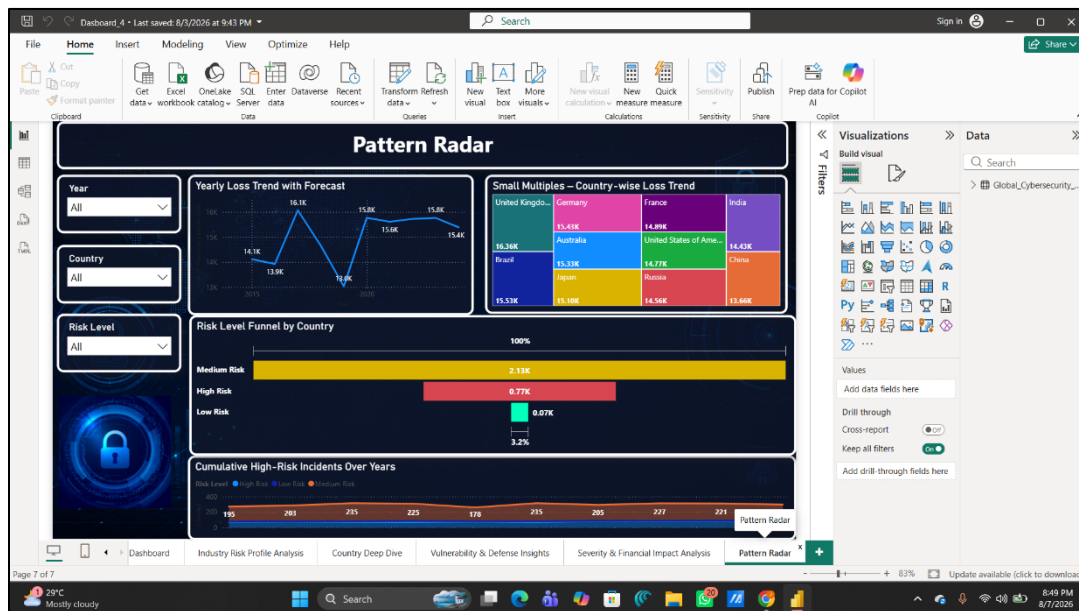
greater cybersecurity risk. It provides a clear geographical comparison of high-risk incident concentration.

Outcome: This dashboard provides valuable insights into cybersecurity severity and financial impact by analyzing high-risk incidents, attack sources, industries, countries, and resolution performance. It helps organizations identify critical risk areas, prioritize high-impact threats, improve incident response strategies, and reduce potential financial losses caused by severe cyber attacks.

Pattern Radar

The **Pattern Radar Dashboard** provides a comprehensive analysis of cybersecurity patterns, financial loss trends, risk-level distribution, and high-risk incident behavior over time. It focuses on identifying recurring patterns and changes in cybersecurity incidents by analyzing yearly financial losses, country-wise loss trends, risk-level distribution, and cumulative high-risk incidents. The dashboard enables users to understand how cyber threats and their financial impact evolve over time and across different countries. By identifying significant patterns and emerging risk trends, organizations can improve threat monitoring, strengthen preventive strategies, and make data-driven cybersecurity

decisions.



Year(Slicer)



The **Year slicer** allows users to filter the dashboard based on a specific year. This helps analyze cybersecurity patterns and financial loss trends for selected time periods and makes it easier to compare changes in cyber risk over the years.

Country(Slicer)



The **Country slicer** filters the dashboard based on a selected country. It allows users to examine country-specific financial loss

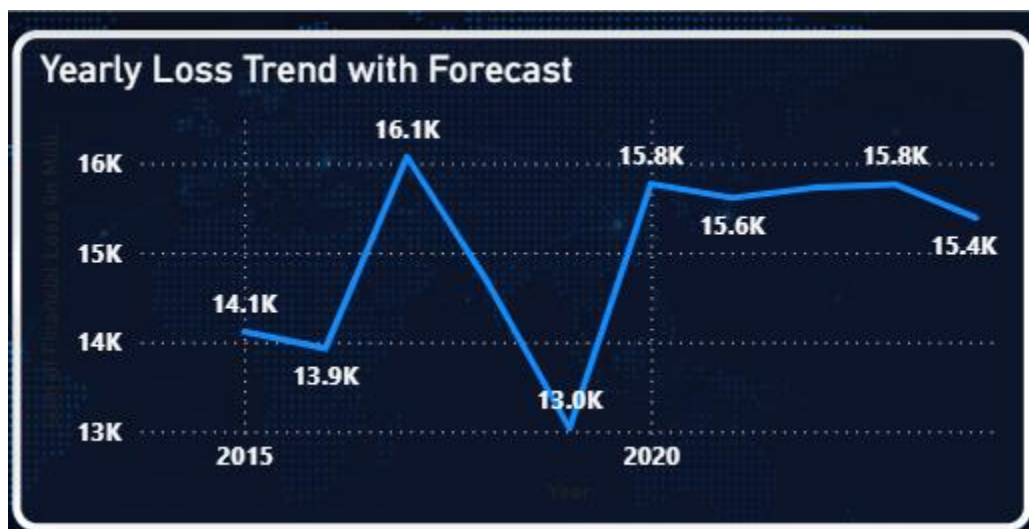
patterns and understand how cybersecurity impact varies across different geographical regions.

Risk Level (Slider)

A dark blue rectangular UI element with rounded corners. At the top, it says "Risk Level" in white. Below it is a white dropdown menu with the word "All" and a downward arrow icon on the right.

The **Risk Level slicer** allows users to filter incidents according to Low, Medium, and High risk categories. This helps users focus on specific severity levels and identify how different risk categories contribute to overall cybersecurity patterns.

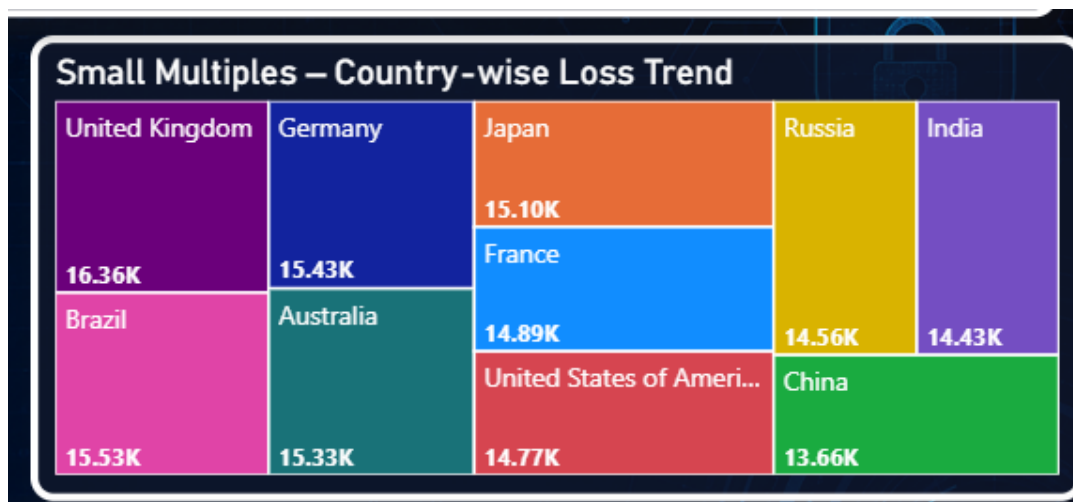
Yearly Loss Trend with Forecast (Line Chart)



The **Line Chart** displays the yearly trend of total financial losses caused by cybersecurity incidents. It helps users identify increases, decreases, and fluctuations in financial losses over time. The forecast component provides an indication of potential

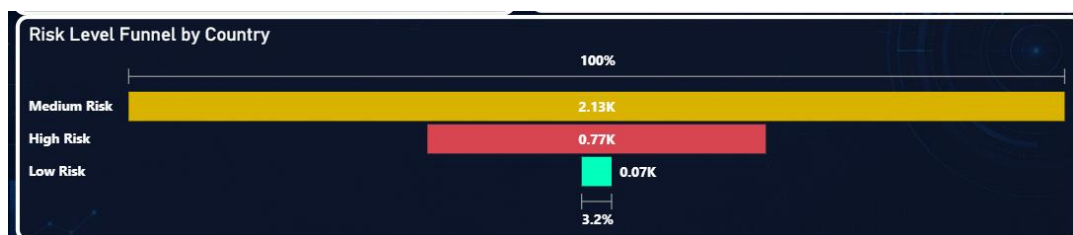
future loss trends, supporting proactive financial and cybersecurity planning.

Small Multiples – Country-wise Loss Trend



The **Small Multiples visualization** displays financial loss trends separately for different countries. Each section represents a country, making it easier to compare how financial losses vary across geographical regions. This visualization helps identify countries with consistently higher losses as well as countries experiencing significant changes over time.

Risk Level Funnel by Country (Funnel Chart)



The **Funnel Chart** illustrates the distribution of incidents across different risk levels, including Medium Risk, High Risk, and Low Risk. It provides a clear view of the proportion of incidents belonging to each severity category and helps users understand the overall risk composition of cybersecurity incidents.

Cumulative High-Risk Incidents Over Years (Area/Line Chart)



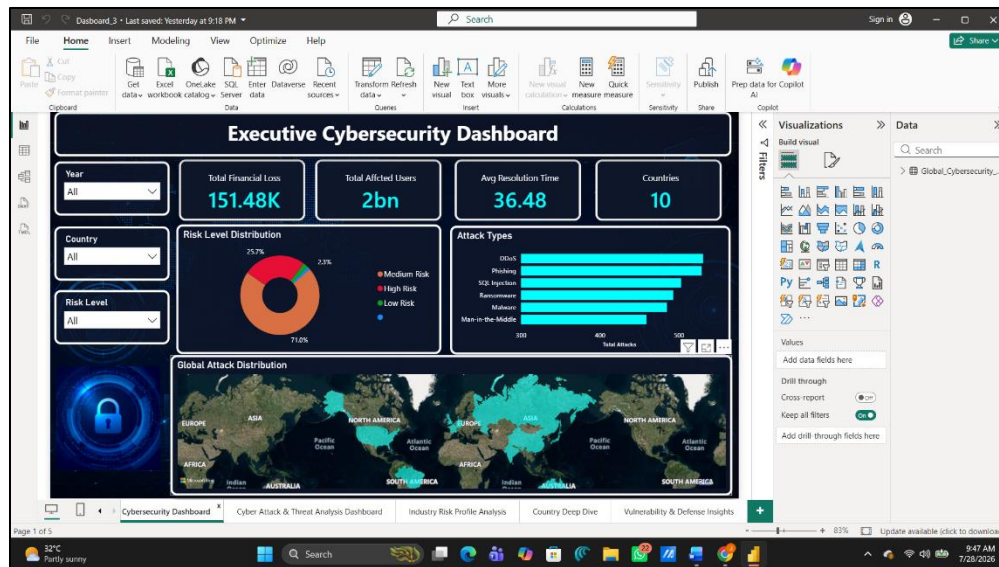
The **Cumulative High-Risk Incidents Over Years** visualization tracks the growth of high-risk, low-risk, and medium-risk incidents across multiple years. It helps identify long-term cybersecurity patterns and shows whether high-severity incidents are increasing, decreasing, or remaining relatively stable over time. This supports long-term threat monitoring and risk assessment.

Outcome: This dashboard provides valuable insights into cybersecurity patterns by analyzing yearly financial loss trends, country-wise loss behavior, risk-level distribution, and cumulative incidents over time. It helps organizations identify

emerging trends, recognize high-risk patterns, monitor changes in financial impact, and support proactive cybersecurity planning and strategic decision-making.

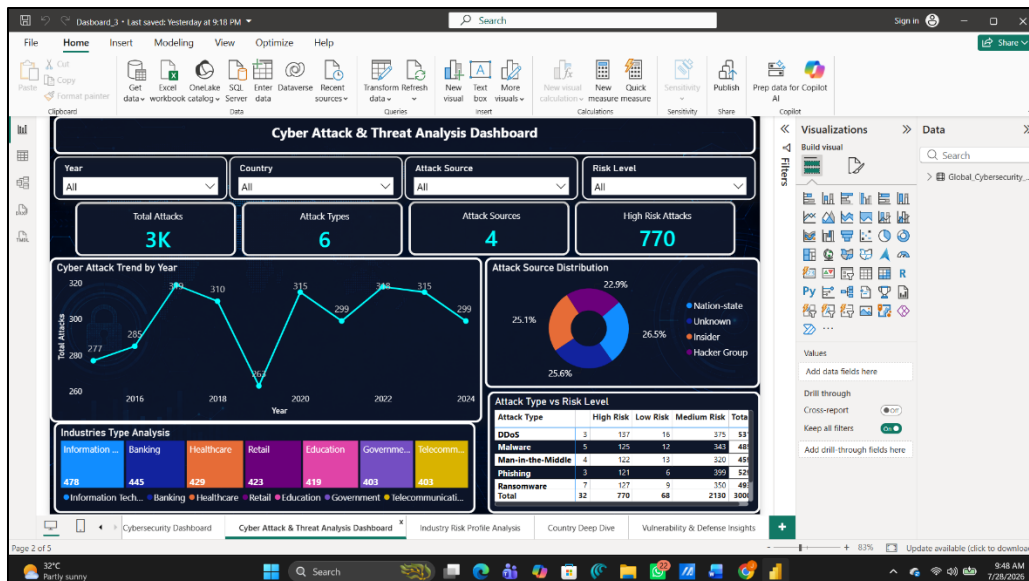
6. Snapshots / Screenshots

Executive Cybersecurity Dashboard



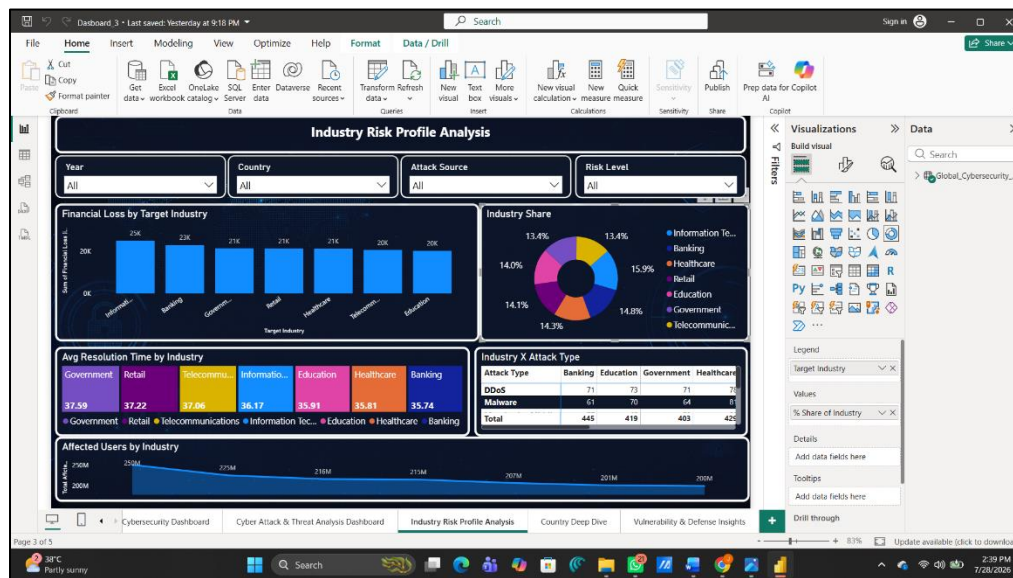
Page 1

Cyber Attack & Threat Analysis Dashboard



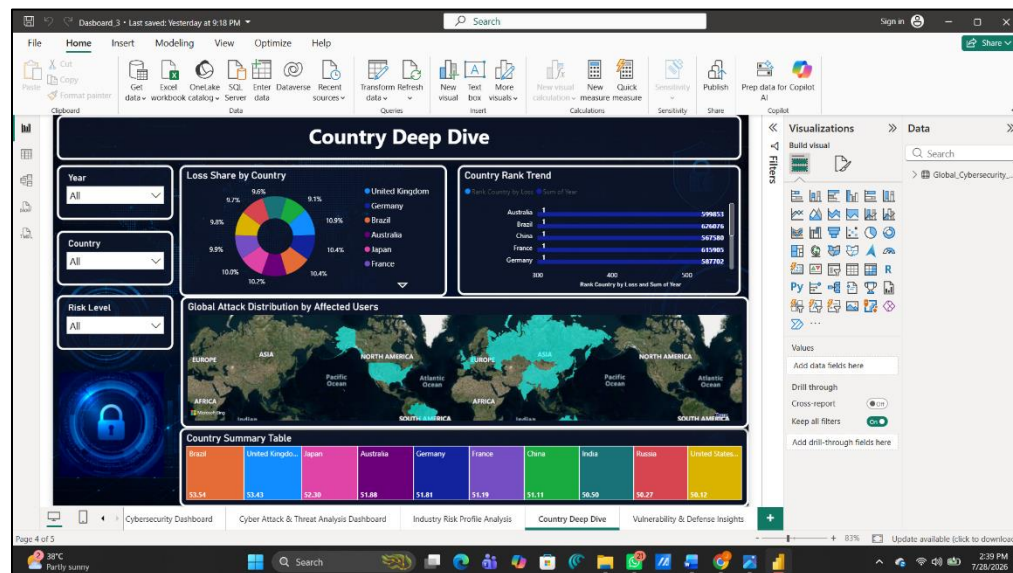
Page 2

Industry Risk Profile Analysis Dashboard



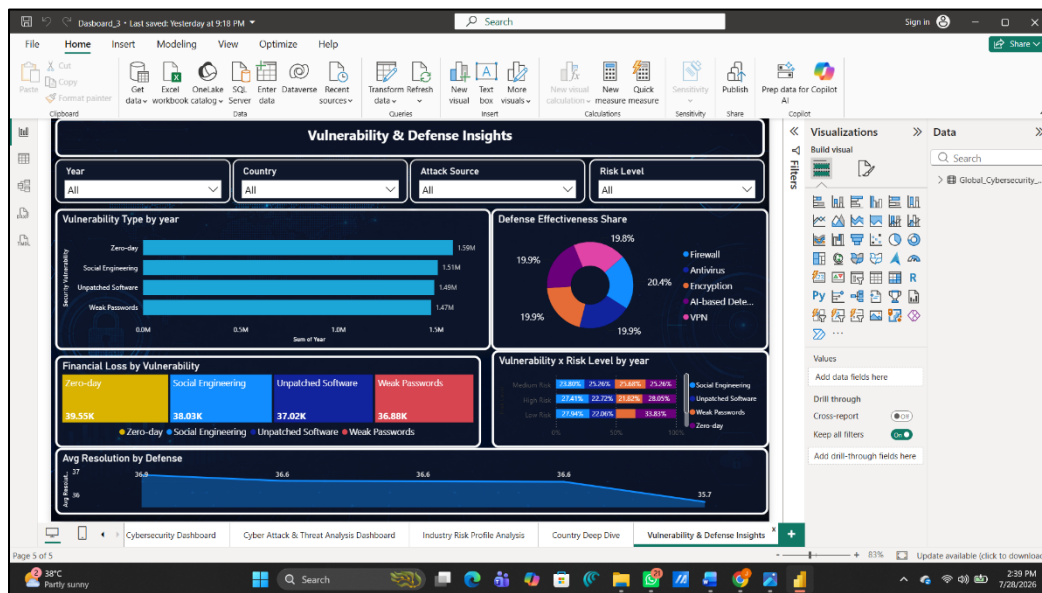
Page 3

Country Deep Dive Dashboard



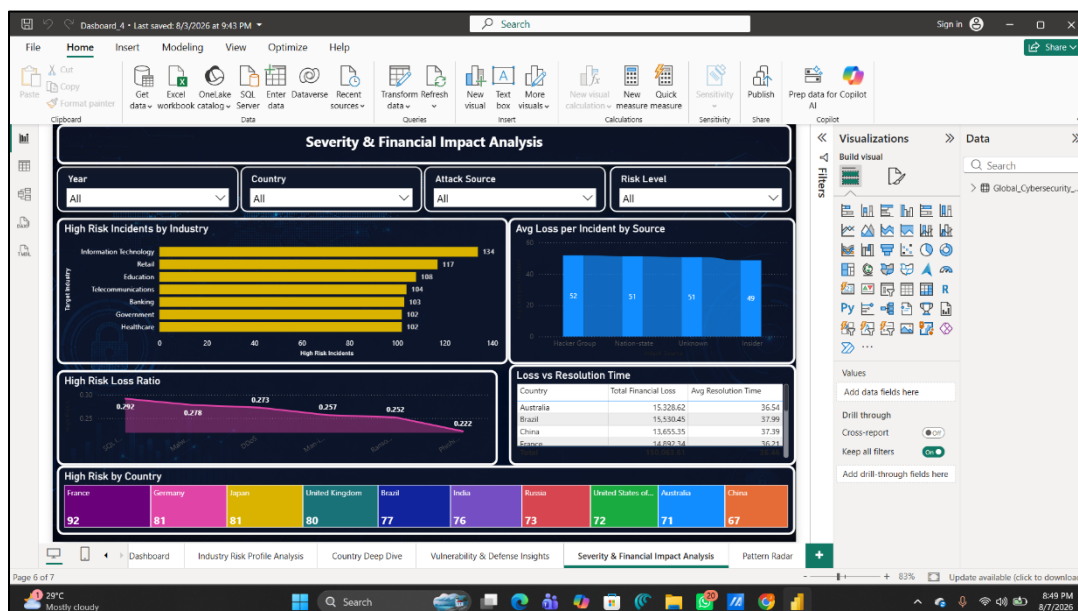
Page 4

Vulnerability & Defense Insights Dashboard



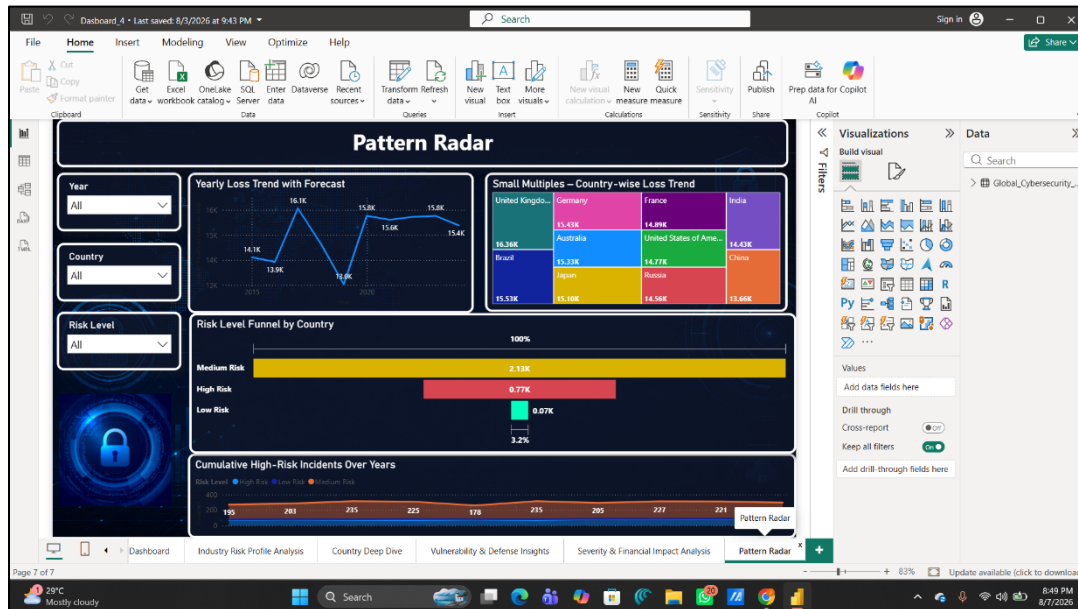
Page 5

Severity & Financial Impact Analysis



Page 6

Pattern Radar



Page 7

7. Challenges Faced

During the project, several technical and analytical challenges were encountered. The dataset required extensive preprocessing due to inconsistent formatting and the need for data cleaning before visualization. Selecting appropriate visualizations for different cybersecurity metrics was another challenge, as each dashboard needed to effectively communicate insights while maintaining readability. Designing interactive dashboards with proper filters, slicers, and navigation required multiple iterations. Performance optimization was also necessary to ensure smooth interaction between visuals. These challenges were addressed through data preprocessing, continuous testing, dashboard optimization, and mentor guidance.

8. Learnings & Skills Acquired

8.1 Technical Skills

During this internship, I gained practical experience in Microsoft Power BI for building professional business intelligence dashboards. I developed skills in data cleaning using Power Query, creating DAX measures, designing interactive visualizations, implementing slicers and filters, building KPI cards, and generating meaningful reports from large datasets. I also learned data modeling techniques and effective dashboard design principles.

8.2 Analytical and Problem-Solving Skills

The project enhanced my ability to analyze large cybersecurity datasets and extract meaningful business insights. I learned to identify trends, compare multiple performance indicators, evaluate risk levels, and solve data quality issues through preprocessing techniques. The internship also strengthened my ability to interpret analytical results and convert raw data into actionable information for decision-making.

8.3 Future Enhancements

The project can be further enhanced by integrating real-time cybersecurity threat feeds instead of static datasets. Machine learning algorithms can be incorporated to predict future cyber attacks and detect anomalies automatically. Additional dashboards can include threat forecasting, AI-based risk prediction, and automated alert generation. Publishing the dashboards through Power BI Service with scheduled data refresh would further improve real-world usability.

8.4 Soft Skills and Team Collaboration

Throughout the internship, I improved several professional skills, including communication, teamwork, time management, presentation skills, and documentation. Working collaboratively with team members helped me understand task allocation, knowledge sharing, and effective problem-solving in a project

environment. Regular mentor guidance also improved my confidence in presenting analytical findings professionally.

8.5 Domain Knowledge and Application

The internship provided valuable knowledge about cybersecurity concepts, including attack types, security vulnerabilities, defense mechanisms, financial impact, and industry risk analysis. I gained a better understanding of how Business Intelligence tools can be applied to cybersecurity data for monitoring threats, identifying vulnerabilities, improving organizational security strategies, and supporting evidence-based business decisions.

9. Testimonials from Team

Janani G K

I contributed to the development of the following Power BI dashboard pages:

- Industry Risk Profile Analysis Dashboard
- Pattern Radar Dashboard

I designed and developed these dashboards by selecting appropriate visualizations, creating DAX measures and KPIs, implementing interactive slicers, and organizing the dashboard layouts for better user experience. I also handled the **cleaning, preprocessing, and preparation of the dataset for analysis**. In addition, I reviewed all dashboard pages and made the necessary changes, corrections, refinements, and improvements to the visualizations, KPIs, slicers, layouts, and overall dashboard design. I ensured consistency, accuracy, proper functionality, and quality across all dashboards and supported the successful completion of the project.

Himanshu Kumar Soni

He contributed to the development of the **Executive Summary Dashboard** by designing KPI cards, charts, visualizations, and interactive slicers to present the overall cybersecurity data in a clear and meaningful manner. He also organized the dashboard layout and ensured that the visualizations effectively communicated the key insights.

Narayani Kantamreddy

She contributed to the development of the **Attack Type Analysis Dashboard** by designing and organizing visualizations to analyze different types of cybersecurity attacks. She worked on charts, KPI elements, and interactive filters to present attack-related insights clearly and effectively.

Bindu Kotnala

She contributed to the development of the **Severity & Financial Impact Analysis Dashboard** by designing and organizing visualizations to analyze the severity of cybersecurity incidents and their financial impact. She worked on appropriate charts, KPI elements, and interactive filters to present the relationship between incident severity and financial losses clearly and effectively. She also supported the data cleaning, final review and testing of the dashboards to ensure proper functionality and presentation.

Nandana N Nair

She contributed to the development of the **Vulnerability & Defense Insights Dashboard** by designing and organizing visualizations related to security vulnerabilities and defense mechanisms. She worked on presenting the analytical insights clearly through appropriate charts, KPIs, and interactive elements.

Gohitha Meghana N

She contributed to the development of the **Country Deep Dive Dashboard** by working on country-level cybersecurity visualizations, charts, KPI elements, and interactive slicers. She also contributed to the cleaning, preprocessing, and preparation of the dataset to ensure that the data was suitable for accurate analysis and visualization.

10. Conclusion

The Cyber Threat Analytics Platform with Incident Visualization System successfully demonstrates how cybersecurity data can be transformed into meaningful business insights using Microsoft Power BI.

By collecting, cleaning, and analyzing cybersecurity incident data, the project provides valuable information about attack trends, financial losses, targeted industries, and defense mechanisms. The interactive dashboards allow organizations to monitor incidents efficiently, identify potential threats quickly, and make informed security decisions.

Overall, this project highlights the importance of data analytics and visualization in strengthening

cybersecurity management and supporting proactive threat detection. It also demonstrates how Power BI can be effectively used to convert raw cybersecurity data into actionable insights for organizations.

11. Acknowledgements

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through this internship have significantly strengthened our analytical thinking, technical proficiency, and business interpretation skills. This experience has enhanced our ability to approach data with a strategic mindset, collaborate effectively within a team environment, and execute projects with professional discipline. Overall, this internship has been a meaningful milestone in our academic and professional journey, reinforcing the value of data-driven decision-making and structured analytical execution in modern business environments.